



TRAMONTANA ICS-01 – THE DISTRIBUTION OF IMPROVED COOKSTOVES TO INDIGENOUS COMMUNITIES ACROSS RURAL INDIA

Tramontana

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Prepared by	Tramontana Asset Management

¹ This date refers to the date the original PDMR was submitted to Verra for Joint Validation and Verification under the previous methodology

CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Audit History.....	7
1.3	Sectoral Scope and Project Type	7
1.4	Project Eligibility	7
1.5	Project Design	8
1.6	Project Proponent	9
1.7	Other Entities Involved in the Project	10
1.8	Ownership.....	10
1.9	Project Start Date	10
1.10	Project Crediting Period	10
1.11	Project Scale and Estimated GHG Emission Reductions or Removals	11
1.12	Description of the Project Activity	11
1.13	Project Location	16
1.14	Conditions Prior to Project Initiation	17
1.15	Compliance with Laws, Statutes and Other Regulatory Frameworks	19
1.16	Double Counting and Participation under Other GHG Programs	19
1.17	Double Claiming, Other Forms of Credit, and Scope 3 Emissions.....	19
1.18	Sustainable Development Contributions	20
1.19	Additional Information Relevant to the Project	31
2	SAFEGUARDS AND STAKEHOLDER ENGAGEMENT	32
2.1	Stakeholder Engagement and Consultation.....	32
2.2	Risks to Stakeholders and the Environment.....	42
2.3	Respect for Human Rights and Equity	45
2.4	Ecosystem Health	48
3	APPLICATION OF METHODOLOGY.....	50
3.1	Title and Reference of Methodology	50
3.2	Applicability of Methodology	50
3.3	Project Boundary	58
3.4	Baseline Scenario	61

3.5	Additionality	66
3.6	Methodology Deviations	67
4	QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS.....	68
4.1	Baseline Emissions	68
4.2	Project Emissions	71
4.3	Leakage Emissions	72
4.4	Estimated GHG Emission Reductions and Carbon Dioxide Removals	72
5	MONITORING	83
5.1	Data and Parameters Available at Validation	83
5.2	Data and Parameters Monitored.....	87
5.3	Monitoring Plan.....	96
	APPENDIX 1	109
	APPENDIX 2	113
	APPENDIX 3	115

1 PROJECT DETAILS

1.1 Summary Description of the Project

A recent Government of India report for 2019-21², concluded that 43.3% of the Indian population use solid fuels for cooking. The majority of these, or 33.4% of the total population which equates to approximately 450 million people, specifically use wood and are located in rural areas. Cooking stoves (or “cookstoves”) form a central component of a family’s life in rural communities, as they are typically used for cooking, room heating and boiling water for drinking and bathing. The traditional stove type used is predominantly a fixed mud clay stove (shown in Figure 1 and Figure 15) and in some cases a three-stone stove (shown in Figure 3). The report¹ estimates that of the Indian population using solid fuels, 95.4% use mud clay stoves. The traditional stoves used are mainly fuelled by unsustainably sourced firewood from local forests. The firewood then often needs to be carried over long distances by headload, as depicted in Figure 1.



Figure 1. Indigenous community member in the project area using a traditional mud clay cookstove and another carrying firewood by headload to her home

1.1.1 Health Implications

The World Health Organisation estimated that over 800,000 people³ in India lost their lives prematurely in 2019 due to household air pollution. This is the largest number of any country. The primary cause is the use of poorly designed traditional inefficient biomass cookstoves in homes, which results in incomplete combustion and large quantities of toxic pollutants being emitted into the immediate environment, in the form of smoke, soot and other particulate matter (PM).

² <https://www.dhsprogram.com/pubs/pdf/FR375/FR375.pdf>

³ <https://www.who.int/data/gho/data/indicators/indicator-details/GHO/household-air-pollution-attributable-deaths>

1.1.2 Environmental Implications

The World Bank⁴ estimates that traditional cookstoves have a low thermal efficiency, in the range 8-15% depending on their method of use. This results in significant demand for firewood, which causes large quantities of greenhouse gas (GHG) emissions and widespread deforestation and degradation of local forested areas.

1.1.3 Social Implications

Typical stove related tasks such as sourcing firewood, cooking and cleaning are predominantly undertaken by women. Moreover, the considerable time they spend collecting and carrying heavy bundles of firewood over long distances has resulted in physical health problems and exposure to dangers, such as wild animal attacks.

1.1.4 Economic Implications

Stove related tasks in total consume between an estimated 6 and 8 hours per day on average (refer to section 3.4). It also forms the primary role of female members in the communities. This minimises the time they have for undertaking more productive activities such as paid employment, agricultural work or spending time in education or training.

1.1.5 Improving The Situation

The Tramontana ICS-01 project, certified under the Verra Verified Carbon Standard (VCS), aims to use carbon finance to distribute up to 400,000 safe energy-efficient heavy-duty improved cookstoves (ICS). The ICS, which are manufactured locally in India, will be provided free of cost to households across multiple states of the country. The project specifically targets low income and socially disadvantaged indigenous communities, who are generally considered to be the most marginalised and underrepresented in Indian society. These communities have been unable to afford or access cleaner cookstoves primarily due to their economic circumstances and remote locations. The project aims to directly and immediately enhance the health, well-being and livelihoods of the communities, and particularly of women and children who are the most affected.

1.1.6 Implementation Plan

Tramontana has identified multiple target areas across several states of India which suffer from local deforestation and are primarily populated with indigenous communities. Almost all of the families in the project area own a Below Poverty Line (BPL) or ration support card issued by the government. This card indicates that the whole family has a maximum monthly income of Rs 10,000 (or equivalent to approximately USD3.50 per day) (see section 1.14).

A distribution plan covering multiple instances under a grouped project has been developed. The initial instance is located in the state of Andhra Pradesh, from which the project can expand to other states including Odisha, Telangana and Chhattisgarh. The first project instance will distribute

⁴<https://documents1.worldbank.org/curated/en/600391468771661997/pdf/ESM2760PAPER0Impact0of0Energy.pdf>

approximately 50,000 ICS after which multiple instances are planned according to the schedule presented in Table 1. Further information on the project location is provided in section 1.13.

The total GHG emissions reductions estimated from the project activity over the first crediting period is 5,381,237 tonnes of CO₂ equivalent (tCO₂e), with an annual average of 768,748 tCO₂e (see section 4.4).

1.1.7 Implementation Status

Over the first monitoring period between 21-January-2023 and 23-August-2023, 46,160 ICS were installed in households. Information on ICS installation and the beneficiaries is recorded in the distribution database, which is managed by the Project Proponent.



Figure 2. Tramontana site visit to oversee ICS distribution and a recipient signing to complete an ICS survey via the Tramontana ICS software application

Table 1. Planned distribution schedule of ICS across the project area corresponding to instances under the grouped project

Instance	Planned ICS Distribution
Region 1	50,000
Region 2	100,000
Region 3	100,000
Region 4	100,000
Region 5	50,000

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Joint Validation + Verification	21-January-2023 – 23-August-2023	VCS	Earthood Services Private Limited	0.586
Requantification + Methodology Change	21-January-2023 – 23-August-2023	VCS	Carbon Check (India) Private Limited	0.586

1.3 Sectoral Scope and Project Type

Sectoral scope ⁵	03 – Energy Demand
Project activity type	Project Type: II – Energy Efficiency Improvement Project Grouped Project: Yes

1.4 Project Eligibility

1.4.1 General eligibility

The project involves energy-efficient cookstove distribution which falls under the sectoral scope 03 - *Energy demand* and is supported by a methodology approved under the VCS Programme. The project activity does not fall under any of the categories of excluded projects stated in the VCS Standard Table 2.1. Therefore it is eligible under the scope of the VCS Programme.

The project has followed the most recent version of the VM0050 methodology version 1.0 published in October 2024. It was successfully listed in the Verra registry on the 16th June 2023. Note that since listing, the project has been updated from the previous methodology, VMR0006 v1.2.

In line with section 3.6.8 of the VCS Standard v.4.7⁶ which states that “the project proponent shall include in a singular project all project activity instances within ten kilometres of another instance of the same project activity and with the same project proponent”, all project activity instances included to date and in the future within close proximity (ten kilometres) of one another will form part of this single project, and will not be separated into a new project.

⁵ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

⁶ <https://verra.org/wp-content/uploads/2024/04/VCS-Standard-v4.7-FINAL-4.15.24.pdf>

1.5 Project Design

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

1.5.1 Grouped project design

For the inclusion of new project activity instances, the Project Proponent must ensure that it meets the following eligibility criteria from the Verra VCS Standard, version 4.7⁵

Eligibility criteria	Compliance by new project instances
Meet the applicability conditions set out in the methodology applied to the project	All new instances under the grouped project will meet the conditions of the methodology as specified in section 3.2
Use the technologies or measures specified in the project description	The technology type employed is an energy efficient ICS to replace the traditional inefficient cookstoves used by households. Information on the ICS technology used for all instances is provided in the form of manufacturer product specifications (see section 1.12), independent third-party certification for ICS thermal efficiency (see Appendix 2 and 3) and through distribution data/photos.
Apply the technologies or measures in the same manner as specified in the project description	Only thermally efficient ICS will be distributed to households under project instances, and no other technologies will be adopted

Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area	All new project instances will remain within the geographical boundaries of India and will therefore share some similar characteristics. However, given project instances corresponds to different areas, the Project Proponent has decided to implement separate baseline campaigns for each instance to ensure a more accurate representation. The baseline scenario, including for the first instance, is discussed in section 3.4. Baseline data for subsequent instances is provided during verification.
Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area	All new project instances will have additionality characteristics consistent with the initial instance, and this will be demonstrated using the activity methodology below. Step 1: Regulatory Surplus Any new project instances included will not be mandated by any law, government policy or regulation in the host country Step 2: Positive List The inclusion of new project instances will comply with the positive list under the methodology (see section 3.5 for details) Step 3: Project Method Not applicable

1.6 Project Proponent

Organization name	Tramontana Impact Limited (subsidiary of Tramontana Asset Management)
Contact person	Dr Ramanan M. Loganathan
Title	Director
Address	2nd Floor, Gaspé House, 66-72 Esplanade, St. Helier, Jersey, JE1 1GH
Telephone	+44 (0)203 773 0160
Email	impact@tramontana.co.uk

1.7 Other Entities Involved in the Project

Organization name	Tramontana Sustainable Commodities Limited (subsidiary of Tramontana Asset Management and formerly called CropZone Agro Forestry Private Limited)
Role in the project	Local Implementation
Contact person	Mr Saibaba Abburi
Title	Director
Address	Road No. 36, Jubilee Hills, Hyderabad, Telangana-500033, India
Telephone	+91 98 4960 2999
Email	saibaba@cropzone.in

1.8 Ownership

The project ownership is with the Project Proponent. All rights to carbon credits (or Verified Carbon Units under Verra) generated from project instances will be with the Project Proponent by means of a standardised licence agreement physically signed by the beneficiary of each ICS at the ICS distribution stage.

1.9 Project Start Date

Project start date	21-January-2023
Justification	Date of installation of the first ICS under the grouped project

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	21-January-2023 to 20-January-2030

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☐ < 300,000 tCO₂e/year (project)

☒ ≥ 300,000 tCO₂e/year (large project)

The following are annual emissions reduction estimates from the project activity over the first crediting period. Details of the assumptions and calculations are provided in section 5.

Calendar Year of Crediting Period	Estimated GHG emissions reductions (tCO ₂ e)
21-January-2023 to 31-December-2023	109,674
01-January-2024 to 31-December-2024	493,801
01-January-2025 to 31-December-2025	917,498
01-January-2026 to 31-December-2026	1,040,856
01-January-2027 to 31-December-2027	989,540
01-January-2028 to 31-December-2028	928,264
01-January-2029 to 31-December-2029	854,369
01-January-2030 to 20-January-2030	47,235
Total estimated ERs (first crediting period)	5,381,237
Total number of years	7
Average annual ERs	768,748

1.12 Description of the Project Activity

Household air pollution from traditional inefficient biomass cookstoves in rural homes is a leading cause of premature deaths in India (see section 1.1). Despite recent government efforts to promote cleaner cooking fuels (through schemes such as PMUY), challenges have occurred including uptake owing to high fuel costs (e.g. for LPG refills) and supply chain issues with fuel distribution to rural areas. The project activity enables the distribution of Tramontana ICS (Figure

4 and Figure 5) directly to individual households in rural areas across multiple states in India. These ICS are provided free of cost, using carbon financing as the means to overcome a key hurdle for their adoption. They will replace traditional inefficient cookstoves, typically fixed mud clay stoves (made using mud, dung and brick or stone) (see Figure 1 and Figure 15) or three-stone stoves (see Figure 3), which use woody biomass as a fuel and have been utilised by the communities for generations.



Figure 3. A traditional three-stone cookstove and a community member in the project area using this stove type

The project is aimed primarily at households who are marginalised from a socio-economic standpoint. The majority of households belong to communities that have been statutorily recognised (in the Indian Constitution and Law) as Scheduled Castes or Scheduled Tribes (defined in Article 341 and 342 respectively of the Indian Constitution), which are considered the most disadvantaged socio-economic groups in India. The state government often provides documentation allowing these households to prove their community status, which in turn allows them to avail of several targeted welfare schemes such as subsidised food grains and preferential access to education and jobs. Note that evidence of such document is collected from ICS beneficiaries and stored electronically during the distribution phase of the project activity (see section 5.3.1).

The Tramontana ICS to be distributed forms a safe, portable cleaner cooking technology which can be (voluntarily) adopted by the communities and does not require them to make major modifications to their current cooking practices. The ICS has an increased thermal efficiency and completeness of combustion through its energy-efficient design, technology and the materials used. GHG emissions are immediately reduced, and families experience significant improvements in the levels of toxic indoor air pollution. The improved efficiency lowers deforestation/forest degradation (and consequently biodiversity loss) and enhances the daily lives particularly of female family members, who are most involved in carrying out the daily tasks relating to the stove.

A pilot study was conducted starting in January 2022, approximately one year in advance of distribution of the first ICS under the project. The study involved distributing 100 ICS within the project region and continuously monitoring usage and performance. This enabled the Project

Proponent to make necessary ICS design improvements and incorporate feedback from community members. The pilot study also helped validate the robustness of the entire supply chain.



Figure 4. Tramontana ICS CRS-23 with unique serial number, QR code and Tramontana branding

Upon project implementation, as part of the ICS distribution process, the project team directly engage with each household and complete a number of tasks including:

- Provide physical demonstrations on how to use the Tramontana ICS
- Make recommendations on best practices including with fuel preparation (e.g. bark shredding, firewood sizing) and stove maintenance
- Hand over the assigned ICS together with an instructions brochure
- Use the Tramontana ICS software application to record an extensive set of data on the user (as outlined in section 5.3.1)

The technical specifications of the Tramontana ICS distributed under the project activity are outlined in Tables 2–5. These specifications will be used to determine emissions reductions for the project. The first 150,000 stoves distributed under the project activity are planned for ICS model CRS-23 (illustrated in Figure 4). Following R&D and incorporating feedback gathered from beneficiaries in Instances 1 and 2, a more thermally efficient and durable stove has been developed, STAR-25 (presented in Figure 5). It is planned for the STAR-25 ICS to be distributed across project Instances 3, 4 and 5.

Table 2. Technical specifications of the Tramontana ICS model CRS-23

ICS Model	CRS-23
Type	Natural Draft
Thermal Efficiency	36.01%

Height	25.5 cm
Diameter	27 cm
Weight	9.47 kg
Eligible Fuel Type	Wood, Charcoal, Other Biomass
Portable or Fixed	Portable
Colour / Finish	Black & White / Glossy
Expected Lifespan	7 years

Table 3. Material components of the Tramontana ICS model CRS-23

ICS Part	Material	Thickness	Tolerance
Top Plate	Cold Rolled (CR) Steel	0.8 mm	+/- 0.1 mm
Grate	Cast Iron	6 mm	+/- 0.5 mm
Barrel/Combustion Chamber	Ceramic	23 - 65 mm (height variation)	+/- 2 mm
Bottom Ring	CR Steel	1 mm	+/- 0.02 mm
Neck (Outer Jacket)	CR Steel	0.6 mm	+/- 0.02 mm
Neck (Inner Lining)	Stainless Steel 430	0.6 mm	+/- 0.02 mm
Perforated Sleeve	CR Steel	0.6 mm	+/- 0.02 mm
Outer Jacket Cladding	CR Steel	0.6 mm	+/- 0.02 mm
Handles	Mild Steel wire covered with Bakelite	8 - 12 mm	+/- 2 mm
Pan Support	CR Steel	2 mm	+/- 0.02 mm
Legs	CR Steel	1 mm	+/- 0.02 mm
Fuel Stand	Mild Steel	4 mm	+/- 0.08 mm



Figure 5: Tramontana ICS STAR-25 with unique serial number, QR code and Tramontana branding

Table 4. Technical specifications of the Tramontana ICS model STAR-25

ICS Model	STAR-25
Type	Natural Draft
Thermal Efficiency	52.50%
Height	27.6 cm
Diameter	38.6 cm
Weight	8.65kg
Eligible Fuel Type	Wood, Charcoal, Other Biomass
Portable or Fixed	Portable
Colour / Finish	Silver / Glossy
Expected Lifespan	7 years

Table 5. Material components of the Tramontana ICS model STAR-25

ICS Part	Material	Thickness	Tolerance
Top Plate	Stainless Steel (304)	0.8 mm	+/- 0.02 mm
Grate	Cast Iron	6.0 mm	+/- 0.50 mm
Barrel/Combustion Chamber	Ceramic	35.0 mm	+/- 2.00 mm
Air Flow Vent	Stainless Steel (202)	0.6 mm	+/- 0.02 mm
Body	Stainless Steel (202)	0.8 mm	+/- 0.02 mm
Chamber Ring	Stainless Steel (202)	0.8 mm	+/- 0.02 mm
Perforated Sleeve	Stainless Steel (202)	0.6 mm	+/- 0.02 mm
Door Frame	Stainless Steel (202)	0.6 mm	+/- 0.02 mm
Handles	Stainless Steel+ Bakelite	6.5 mm	+/- 0.02 mm
Pan Support	Stainless Steel (202)	2.0 mm	+/- 0.02 mm
Legs	Stainless Steel (202)	2.0 mm	+/- 0.02 mm
Fuel Stand	Mild Steel	4.0 mm	+/- 0.02 mm

1.13 Project Location

All project activity under the grouped project is within the geographical boundaries of India depicted in Figure 6⁷ with approximate coordinates of latitude 21.555808 and longitude 78.746972. A separate KML file is provided describing the project location. The first project instance is within the Paderu division of Alluri Sitarama Raju (ASR) district of Andhra Pradesh state shown in Figure 7⁸, with approximate coordinates of latitude 18.011959 and longitude 82.555030. It is a high elevation region at approximately 1000 metres above sea level.

⁷ <https://www.mapsofindia.com/images2/india-map.jpg>

⁸ <https://www.india.com/news/india/andhra-pradesh-to-have-13-new-districts-from-april-4-check-list-of-total-26-districts-here-5317197>



Figure 6. Project activity area of India delineated by state

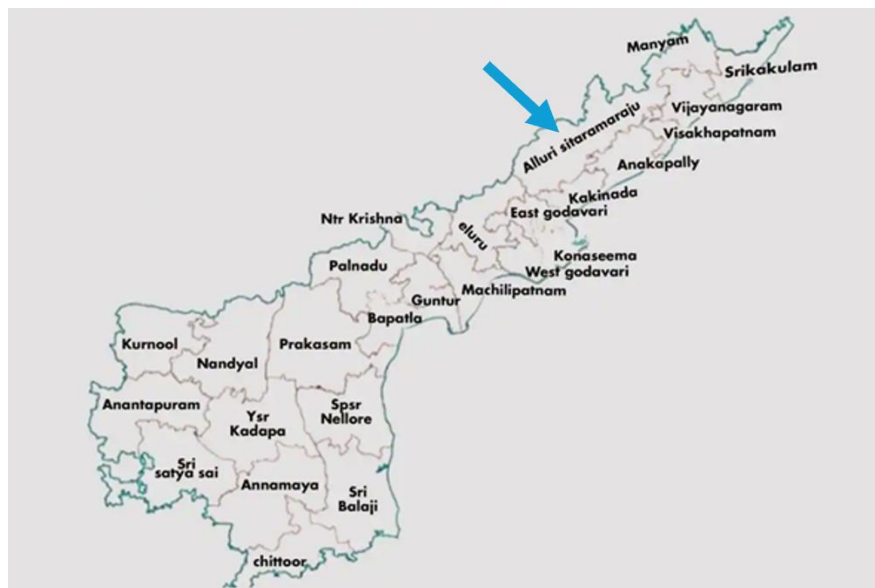


Figure 7. Map of Andhra Pradesh state in India divided by district where the first project activity instance is located (see blue arrow)

1.14 Conditions Prior to Project Initiation

The target communities live in rural areas and are predominantly from indigenous and socio-economically disadvantaged communities. They rely on nature for their survival, and reside in environments ranging from forests to hills with many in difficult-to-access remote areas. The level of education is generally low and many children lack access to secondary schools or drop out early

to help out their family. Almost all families own a Below Poverty Line (BPL)/ration card issued by the state government confirming eligibility to receive food provisions, subsidies etc, which in Andhra Pradesh (and similarly in other states) is only handed to those households with a maximum annual income for the whole family of less than Rs 10,000 per month (or equivalently USD3.50 per day approximately) amongst other conditions⁹. Documentary proof of this, where available, is collected during ICS distribution and recorded in the project distribution database.

The pre-project baseline scenario is the continued use of rudimentary traditional cookstoves by the communities, typically in the form of fixed mud clay stoves (Figure 1 and Figure 15) or three-stone stoves (Figure 3). Usage of these stoves has been commonplace in the project areas for generations, where households burn woody biomass to meet their thermal energy needs including for the purposes of cooking, heating and boiling water. These cookstoves are considered to have a low thermal efficiency, in the range 8-15% depending on their method of use according to the World Bank¹⁰. Further, their poor design results in incomplete combustion which releases significant quantities of toxic air pollutants into the immediate environment, creating a black coating on pots, cooking utensils and walls. Toxic air pollutants produced cause issues such as breathlessness, coughing, headaches, burning eyes and longer term respiratory/retinal issues in the target communities, particularly as users (mainly women) spend long hours in close proximity to the stoves, and in many cases within a confined environment.

The large firewood requirement of traditional stoves results in significant generation of GHG emissions and deforestation/degradation of the local forested areas (see Figure 8). It further leads to considerable time spent collecting firewood, which prevents primarily female family members from undertaking alternative income generating and skill development activities. Given the heavy loads are carried by headload and for many kilometres, it also causes physical health problems as well as exposure to dangers from the local area including wild animal attacks. To gain further insight into the pre-project conditions, baseline surveys are conducted for each project activity instance under the grouped project. Details for the first project instance can be found in section 3.4.

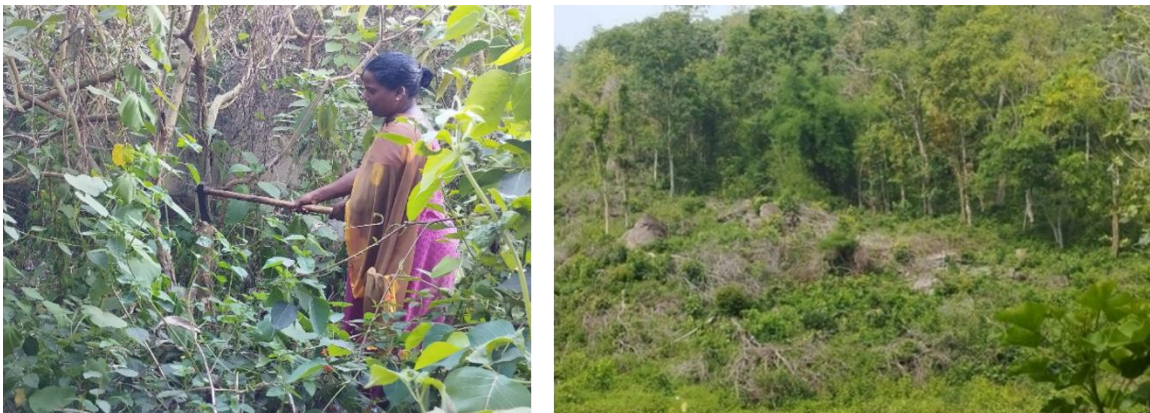


Figure 8: Community member cutting trees to collect firewood to carry home and evidence of recent deforestation in the local area as result of large firewood demand

⁹ <https://civilsupplies.ap.gov.in/Notices.jsp>

¹⁰ <https://documents1.worldbank.org/curated/en/600391468771661997/pdf/ESM2760PAPER0Impact0of0Energy.pdf>

Under the project activity, all beneficiaries will communicate information on their pre-project stove when receiving a new ICS, with responses recorded in the distribution database. There will also be a requirement for demolishing or removing pre-project cookstoves prior to receiving a new ICS in order to minimise any chances of reverting to the previous stove.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

There exists no mandatory law or regulations requiring the introduction or use of ICS in India. The project is undertaken on a voluntary basis by the Project Proponent.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

1.18 Sustainable Development Contributions

The project contributes to environmental, social, economic and technological benefits which lead to the sustainable development of the local area and country as follows:

Environmental Benefits

- Preservation of forest stock and protection of ecosystems, habitats and soil health by reducing the rate of deforestation and forest degradation across the project area.
- Reduction in toxic indoor air pollution from the cleaner combustion characteristics and increased thermal efficiency of the project ICS. This lowers health issues from respiratory and other diseases. Communities also save on medical expenditure and require less time off to look after unwell family members.
- Lower GHG emissions generated by a reduction in the use of non-renewable biomass.
- Less water and effort needed for cleaning utensils and maintenance of the household as a result of the cleaner cooking process and minimised smoke/soot generation.

Social Benefits

- Reduced drudgery and exposure to risks including wild animal attacks, injuries from tree cutting/wood chopping etc. that are prevalent in the project area, as a result of less time needing to be spent sourcing, carrying and processing firewood. These issues were widely reported in baseline surveys for the first project instance (see section 3.4).
- More hours for skill/economic development and children learning owing to reduced time spent sourcing firewood.
- Improvement to gender equality as it is primarily women that carry out the tasks relating to stove usage, firewood collection and cleaning.
- Improved home conditions and quality of life for families by reducing localised indoor air pollution and from the time savings.
- Enhanced safety and reduced burns/injuries for children owing to the compact nature of the Tramontana ICS and less direct exposure to the flame.
- Confidence and comfort as no material change is required to traditional cooking practices.
- Development of IT skills for the local teams employed as a result of training and experience gained under the project activity.

Economic Benefits

- Employment opportunities and skill development for the local communities and economy through the project activity. This includes specifically employing and training

local teams for operational tasks such as distribution, monitoring, maintenance, data analysis and conducting stakeholder meetings.

- More hours available for productive income generating activities and children learning as a result of reduced household work and fuel sourcing.
- Reduced expenditure on firewood (if purchased) due to the increased thermal efficiency of the Tramontana ICS.

Technological Benefits

- Introduction of cleaner technology directly to rural communities.
- Employment of local workers and provision of training and experience in the use of tablets with an Android software application to conduct baseline surveys, distribution and monitoring work.
- Provision of training and experience in IT for local office teams to undertake the tasks around project data compilation and analysis.

Nine of seventeen global UN Sustainable Development Goals (SDGs) have been identified as relevant to this project and are illustrated with black boxes below. Table 6 provides the detailed description of each of these SDGs including the relevant baseline scenario and corresponding contribution from the project activity. Information is also provided on how each SDG contribution will be measured through the project life.



Table 6. Sustainable Development Contributions

SDG Target	SDG Indicator ¹¹	Baseline Scenario	Net Impact on Indicators	Contribution of Project Activity and Monitoring Approach
 NO POVERTY	Project Specific Indicator: Number of people provided access to the basic service of cleaner cooking through the provision of a Tramontana ICS The project specific indicator has been designed to align with SDG target 1.4	The majority of the households in the target communities belong to those that have been statutorily recognised (in the Indian Constitution and Law) as coming from the most disadvantaged socio-economic groups in India. The communities typically own a Below Poverty Line (BPL)/ration government card signifying a maximum family income of Rs 10,000 per month or equivalently USD3.50 per day approximately (see section 1.14). The communities are currently using rudimentary inefficient and highly polluting traditional stoves to meet their thermal energy needs. They are unable to purchase an ICS primarily due to lack of access and affordability, and thus live without the means	Implemented activities to increase	Under the project activity, ICS will be distributed to rural communities that are considered the most disadvantaged socio-economic groups in India. The ICS will provide them with access to clean cooking technology which is considered a basic service for all families/households. In addition, it also reduces the time spent by families in undertaking stove related tasks and therefore allows more time for: income generating activities (e.g. supporting in the main work contributing to the family's income), activities to help with skill development, and community work, which can all improve the family's livelihood. The impact on this SDG can be assessed by first identifying the number of households owning a BPL/ration card through survey responses and documentary proof (where available) in the distribution database. By monitoring the number of people from these community categories that have received and are using an ICS based on

¹¹ www.sdgdata.gov.uk

		to access basic services including clean cooking.		distribution and monitoring database records, the impact on the indicator can be assessed.
 GOOD HEALTH & WELL-BEING	Project Specific Indicator: Proportion of households reporting reduced volumes of smoke inside their homes in addition to general health improvements The project specific indicator has been designed to align with SDG target 3.9	The poor design of traditional stoves used in the target communities results in incomplete combustion of the firewood which releases significant toxic air pollution into the local environment as smoke, soot and other PM. This has contributed to a prevalence of respiratory and retinal issues, and premature deaths in the local communities as users (mainly women) spend long hours in close proximity to the stoves. Further observations on the baseline scenario are given in sections 1.14 and 3.4.	Implemented activities to increase	The improved quality and design of the Tramontana ICS results in an immediate improvement in the levels of smoke/soot generation and toxic indoor air pollution, therefore contributing to improved home conditions for families. Additionally, the portable nature of the ICS provides flexibility in the location of use, which can be for instance elsewhere together with the family or outside. The impact on this SDG can be assessed through monitoring surveys obtaining extensive user feedback on areas including ICS usage/performance, changes in local conditions and impact on general well-being. As part of this extensive survey, beneficiaries are specifically asked: 1. Has the local smoke and air pollution generated changed? 2. Has your health and daily life improved? The responses to these questions are used to measure the impact on the SDG and information is presented in monitoring databases and reports.

				Additionally, the increased efficiency of the ICS means less time is spent sourcing and carrying firewood. This means less of the negative physical and social issues associated with these tasks, and more time with the family and for pursuing alternative tasks.
 QUALITY EDUCATION	Project Specific Indicator 1: Proportion of households reported to be spending more time on education as a result of time savings generated by the project The project specific indicator has been designed to align with SDG target 4.3 Project Specific Indicator 2: Number of adults given specific ICT training under the project The project specific indicator has been	The target communities have limited support, access or even time available to be able to undertake skill development/training or education. Instead, much of the time of family members (including children) is spent helping to carry out tasks for the purposes of survival, such as firewood gathering, cooking/heating, cleaning, agricultural labour for subsistence purposes etc.	Implemented activities to increase	Given the lack of access and opportunities available to community members and their children, improving the time available for education forms an important aspect to the project. The time savings from the reduced firewood collection, as a result of the improved design and efficiency of the ICS, leads to considerably more time available for education. Employed staff are trained under the project to undertake tasks including surveying and data collection using the Tramontana application. Training includes IT skills (using the application) and methods to maintain high data quality/integrity. Each household that becomes a beneficiary of an ICS is educated on areas such as climate change/sustainability, sustainable stove usage practices, ICS design and benefits, firewood preparation, maintenance/repair etc and is provided with

	designed to align with SDG target 4.4			continuous support and engagement throughout the project life. The impact on Project Specific Indicator 1 can be assessed through monitoring survey results identifying the proportion of beneficiaries (mainly their children) that are able to spend more time in education. The impact on Project Specific Indicator 2 can be measured by the number of staff employed under the project and given specific ICT training for their role.
 GENDER EQUALITY	Project Specific Indicator: Proportion of beneficiaries able to spend more time contributing to the main family income, spending time with family and helping others in the community The project specific indicator has been designed to align with SDG target 5.4	It is primarily female members of families that spend a disproportionate amount of time carrying out the daily tasks (unpaid work) around stove preparation and usage. This makes them the most exposed to the negative aspects of stove related tasks including pollution, physical pain and social issues. It also prevents female family members from being able to undertake other activities that generate income for the family, provide skill development, or support the local community.	Implemented activities to increase	Though the ICS project activity alone is unable to alter/balance the role of family members in carrying out the daily tasks, it can drastically reduce the negative impacts associated with firewood collection/preparation and stove usage which is undertaken mainly by the female members. This includes issues such as continuous physical pain which have been experienced during collection of firewood. The reduced negative impact is achieved as result of the improved quality, design and efficiency of the Tramontana ICS, and via the education provided on sustainable practices. The impact on the Indicator is assessed during monitoring surveys. Women are asked:

				1. Has time been saved by using the ICS? 2. If yes, how do you spend that time? The responses to these questions are used to measure the impact on the SDG and information is presented in monitoring reports. In addition, the SDG impact here is also assessed through the number of female members recorded as employed and trained as part of the project activity. This is tracked through employment records and contracts which are stored
 AFFORDABLE AND CLEAN ENERGY	7.1.2 Proportion of the population with primary reliance on clean fuels and technology	There are approximately 450 million people in India that still have a continued reliance on solid fuels (mainly firewood) as a cooking fuel (see section 1.1) and do not have access to cleaner energy sources. Across the project area usage of rudimentary traditional inefficient cookstoves by the communities has been commonplace for generations. Households under the project belong to communities that have been statutorily recognised as suffering from socio-economic	Implemented activities to increase	The Project Proponent supporting this project is an international company based outside of the project region. Through their investment into distributing high quality, efficient ICS free of cost, the target communities can directly and immediately experience the benefits of using cleaner energy/technology and without the need to significantly change their daily practices. The free of cost nature is particularly advantageous for the poor marginalised communities, who would otherwise not be able to afford cleaner technologies. The impact on this SDG can be assessed by multiplying the number of households that are using an ICS by the household size, to

		deprivation, and have been unable to afford any cleaner forms of energy		determine the total number of people under the project that have switched and are benefiting from cleaner energy/technology. The household size is determined at the stove distribution stage and the number of stoves operating can be identified from monitoring survey results for the period.
 DECENT WORK AND ECONOMIC GROWTH	Project Specific Indicator: Number of people employed directly and indirectly as a result of the project The project specific indicator has been designed to align with SDG target 8.5	The marginalised target communities have limited access to skilled jobs and development opportunities given their remote locations, the low-quality of resources/infrastructure available to them, and the minimal outside support. Most people work in subsistence for their families in areas such as agriculture and have very little income, as confirmed by the Below Poverty Line (BPL)/ration card issued to them by the state government.	Implemented activities to increase	The project activity has resulted in new jobs, training and skill development for local people. Specifically, local field and office staff have been employed and supported with extensive training, including: gaining experience using an Android software application on tablets, data entry/acquisition and analysis, and in event set up and hosting for stakeholder meetings and community engagements. In addition, opportunities for coordinator roles have been given to more experienced individuals. The project promotes decent work conditions for all by ensuring continuous support, fair wages and benefits, training, safety, and opportunities for career development. This can lead to a strong positive impact on the livelihoods of local workers and their families, and greater social and economic development/growth in the communities. Outside of the local project areas, many more people are employed (both directly and indirectly) to support the project activity,

				including for manufacturing, transport and logistics etc All these service providers are located within the project boundary of India and thus the project activity contributes to national development and growth. The impact on this SDG can be assessed by observing the records of all persons employed (full-time and part time) under the project activity. Employment information is also obtained from outside service providers where possible. In addition to employment data, other information is collected such as attendance records for all training sessions, number of tablets deployed, stakeholder event details including local speakers and organising staff.
 RESPONSIBLE CONSUMPTION AND PRODUCTION	Project Specific Indicator: Total mass of firewood saved as a result of the project activity The project specific indicator has been designed to align with SDG target 12.2	The unsustainable practices employed across the project area and low thermal efficiency of traditional baseline cookstoves (in the range 8-15% as discussed in section 1.14) has resulted in the overuse of non-renewable biomass, thereby depleting precious local natural resources including forests (see for instance Figure 8). As these practices are widespread across the project area, the negative	Implemented activities to increase	The Tramontana ICS distributed across the project region has an improved design/durability and significantly higher thermal efficiency as compared with the baseline stove. This directly results in a reduction in the consumption of firewood and leads to less time needing to be spent carrying out stove-related tasks including firewood sourcing, carrying and preparation work. Local communities are also educated to adopt sustainable practices and maximise resource efficiency going forward.

		impact on local resources is significant.		The impact on this SDG can be assessed by calculating the total mass of firewood saved through the project activity. This is determined by multiplying the firewood consumption reduction per stove over an operating period by the number of operating stoves observed from monitoring surveys.
 CLIMATE ACTION	13.2.2 Total greenhouse gas emissions per year	Rudimentary traditional cookstoves which are poorly designed and considered to have a low thermal efficiency are widely used across the project area. This results in significant amount of GHG emissions being generated.	Implemented activities to decrease	The replacement of traditional cookstoves with the efficient Tramontana ICS through the project activity will result in an immediate reduction in GHG emissions (and localised air pollution). GHG emissions reductions will occur throughout the lifetime of the project. As an ex-ante estimate based on calculations in section 4, the annual GHG emissions reductions over the first crediting period is 768,748 tCO₂e. The impact on this SDG can be assessed by calculating and verifying the emissions reductions that occur from the operational ICS in each monitoring period. As this is the primary task behind the generation of carbon credits (or Verified Carbon Units under Verra), the SDG does not require any additional monitoring to that already undertaken as part of monitoring/verification under the project activity.

 LIFE ON LAND	Project Specific Indicator: Forest area conserved due to reduced firewood consumption from project implementation The project specific indicator has been designed to align with SDG target 15.1	The firewood used by cookstoves across the project area is typically sourced by family members directly from forested areas (3.31km away on average in the first project instance as indicated in section 3.4). It is obtained by collecting fallen wood or cutting down trees in the forest, sorting the wood and then carrying it home by headload and preparing it for fuel. The firewood obtained is largely non-renewable (see section 3.2), thereby contributing to deforestation and forest degradation across the region (as shown in Figure 8). This has been occurring over generations with little effort to replenish the land through sustainable activities such as replanting.	Implemented activities to increase	The reduction in firewood consumption by the Tramontana ICS as a result of its higher thermal efficiency, leads to a meaningful increase in the preservation of forest stock, associated ecosystems, wildlife habitats and soil health through the reduced rate of deforestation and forest degradation. Local communities are also educated in stakeholder meetings and various community engagements on sustainable practices and the importance of its adoption to minimise critical forest loss. The impact on this SDG can be assessed by calculating the total mass of firewood saved through the project activity and subsequently estimating the amount of forested area preserved using average tree dimensions and spacings. As with SDG 12, the mass of firewood saved is obtained from the product of firewood consumption reduction per stove over a period of operation and number of operating stoves from monitoring survey results.
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1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage in line with the methodology (see section 4).

1.19.2 Commercially Sensitive Information

All information for the purposes of project description and project implementation is included in this report which is made publicly available.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	Pre-baseline surveys are conducted across an entire instance region to identify all eligible households. Subsequently multiple beneficiary stakeholder meetings are held to further engage with eligible ICS beneficiaries and convey more detailed information. Other project stakeholders are directly employed or partnered with at the design and planning phase of a project instance. <u>Primary stakeholders:</u> local families receiving a Tramontana ICS, local employees such as field staff, local administrators (e.g. District Collectors), NGOs <u>Secondary stakeholders:</u> Outside of the local project areas, additional people are employed to support the project activities including manufacturing, transportation etc.
Legal or customary tenure/access rights	As part of the project activities, close engagement with the local communities is facilitated to identify any potential concerns and legal conflicts in relation to the project. By means of access to a local project representative or through a designated phone number, stakeholders can raise any concerns or questions they have throughout the project. To date, no conflicts or issues have been raised. The project does not interfere with any legal or customary rights to territories and resources held by project stakeholders.
Stakeholder diversity and changes over time	The ICS beneficiaries under the project live in rural areas and are from indigenous and socio-economically disadvantaged communities. The vast majority are represented by the Scheduled Tribe/Adivasi or Scheduled Caste/Dalit community categories (discussed in section 1.12). The makeup of the indigenous people within a project instance is similar and all follow similar customs and practices. This is not expected to change over time.

	As part of the project, local staff from the project region are employed and trained to undertake the activities on-site and engage with the communities. All primary and secondary stakeholders involved in the project are located in India.
Expected changes in well-being	The project aims to have a positive impact on stakeholder wellbeing. The social and economic benefits of the project are outlined in detail in Section 1.18 and are briefly listed here: • Reduced drudgery and exposure to risks including wild animal attacks, injuries from tree cutting/wood chopping • More hours available for productive income generating activities and schooling • Improved home conditions and quality of life for families • Enhanced safety and reduced burns/injuries • Development of IT skills for the local teams employed • Employment opportunities
Location of stakeholders	ICS beneficiaries under the first project instance are located in the Paderu division of ASR district in Andhra Pradesh state in India. The precise location of each beneficiary is recorded during ICS installation and is presented in the distribution database. Outside of the local project areas, the people employed to support the project activities (e.g. manufacturing, transport) are all located within the project boundary of India.
Location of resources	Many of the project communities undertake agricultural work on land which has been provided to them (with ownership) by the government. This land is typically in close proximity to the beneficiary's home. The project does not interfere with such territories of the stakeholders.

2.1.2 Stakeholder Consultation and Ongoing Communication

The objective of stakeholder consultation meetings is to present the Tramontana ICS-01 project to the local communities and gather feedback. This mainly involves highlighting the project

objectives, the key benefits of using the ICS device, the stakeholders involved, as well as direct engagement and undertaking question/answer sessions with the communities.

The ICS project is designed to distribute 400,000 ICS across multiple states of India, as a grouped project with separate project instances. Multiple stakeholder consultations will be conducted for each project instance to ensure engagement with the communities in each area.



Figure 9. Newspaper advertisement for the first stakeholder meeting of first project instance held on 14th November 2022

Date of stakeholder consultation	The stakeholder consultations for the first instance comprising of 11 mandals across the Alluri Sitarama Raju (ASR) district of Andhra Pradesh mainly occurred between 14th November 2022 and 4th December 2022. On average two meetings were conducted per mandal which ensured a uniform and extensive consultation with community stakeholders. The schedule followed for the first project instance, which included a total of 2231 attendees, is outlined in Appendix 1. Minutes and attendance sheets were taken for all meetings and are provided to the VVB and Verra. Figure 11 illustrates a newspaper article discussing one of the stakeholder meetings conducted. Meetings with a similar format and structure will be undertaken across all project instances in the future.
Stakeholder engagement process	Prior to stakeholder meetings, pre-baseline assessments are carried out to identify potential eligible households in villages. Once an area has been scoped out and the preliminary data has

been collected, the project team can proceed with organising stakeholder meetings and baseline surveys.

All potential beneficiaries of an ICS within a mandal are invited to attend in-person stakeholder meetings, which are advertised in advance through newspapers (example shown in Figure 9), pamphlets and/or word of mouth (e.g. through local community leaders). At each meeting, banners in the local language containing information on the project and the Tramontana ICS are displayed around the venue. Every attendee is provided with a physical brochure containing similar information (presented in Figure 10), and also offered refreshments during the meeting.

Several special guests are invited to the meetings such as MPs, District Collectors and Community Presidents. These guests have a strong influence in the local area owing to their position of responsibility, knowledge of the local community and good understanding of the challenges faced by the people.

All stakeholder meetings are conducted following local customs and traditions. The meetings are held in the local language, which for the first project instance in Andhra Pradesh is Telugu.

In each meeting the Project Proponent and other local stakeholders are first introduced by a member of the team. Then a detailed presentation is given on the project, including the main objectives and a comprehensive (non-technical) overview of the ICS including highlighting the social (particularly for women), economic, and environmental benefits of adoption. Furthermore, an outline is given of the distribution/monitoring phases of the project and the underlying carbon credit-based financing. One other aspect of the presentation is the emphasis on the responsibilities of all households with regards to correct and continued ICS use, together with the support recipients will receive regarding repairs and replacements.

Following the main presentation, each special guest in turn gives a speech providing their own insight into the project and impact on the local community. Finally, attendees are given the opportunity to ask questions or share any concerns directly with the project team through a microphone handed to them (depicted in Figure 12).

The size of each stakeholder meeting is dependent on the mandal population and response rate. A full list of all attendees at each meeting is recorded, including their signatures, contact number and address. In addition, the minutes of every meeting

	including stakeholder questions and the responses are recorded and documented (in English). All information recorded from meetings including detailed minutes, attendance sheets, stakeholder questions and photos are made available.
Consultation outcome	The stakeholder meetings for the first project instance were all successfully conducted. The project gained a very positive reception and overwhelming support from the local communities across all 11 mandals. All invited special guests were keen advocates of the project activity, communicating their strong support and highlighting the importance of such an initiative for marginalised and underrepresented communities. Many reiterated the message that the Project Proponent is the first ever organisation to provide significant support to their local community and that the improved cooking technology will be life changing. Attendees were given the opportunity to ask questions or raise their concerns during (and after) all stakeholder meetings. Some examples of questions raised by attendees are detailed in Table 7. No negative comments were made by attendees with regards to the project activity, and thus no changes were required to the general project design or operation.
Ongoing communication	People from the local area are employed as field staff and project coordinators. This enables them to connect properly with the local communities and understand their needs during the project life. All concerns or questions that occur following a stakeholder meeting, can be directly reported to a local mandal representative or voiced through a dedicated phone number. These formats are suitable given many local people in the communities are illiterate and unable to express their feedback in written format. All points raised will be added by the team to a local grievance register. The information in the register is reviewed on a regular basis by the Project Proponent, with all concerns addressed and incorporated into the project on a merit basis.

	Regular visits (at least once annually) are made by local project representatives to the villages in each area through the project life.
Stakeholder input	All questions raised and corresponding answers given by the project team during stakeholder meetings are recorded in the form of minutes. All minutes and attendance sheets for the first project instance are provided to the VVB and Verra. Some of the most commonly asked questions for the first project instance are given in Table 7. The project team also made themselves available after all meetings for any further discussion points to ensure all attendees were satisfied. No changes were required to be made to the project design as a result of the questions/discussions. Any key points raised from future stakeholder meetings will be recorded and if justified incorporated into the project design.

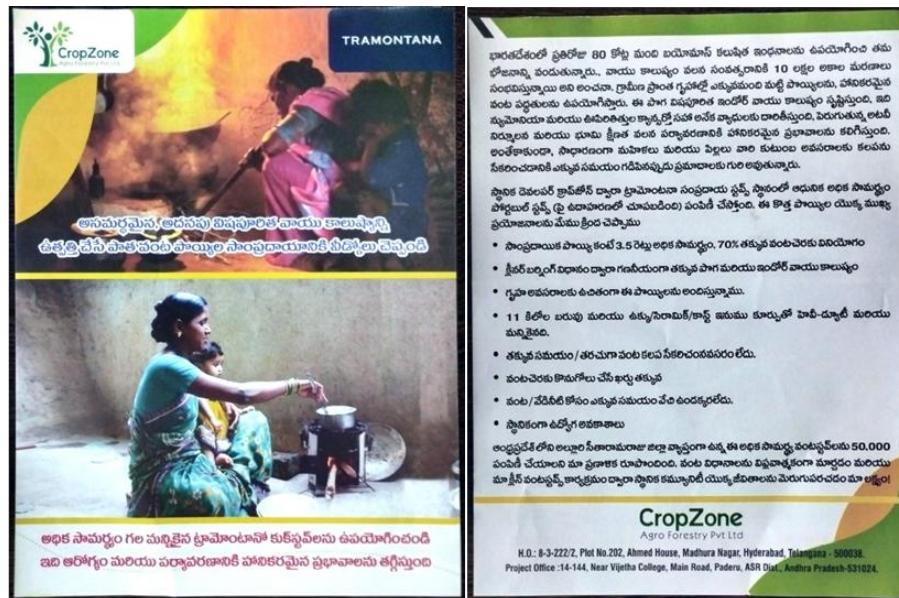


Figure 10. Brochure in the local language distributed at stakeholder meetings providing information on the project and ICS including its benefits





Figure 12. Stakeholder meetings depicting a Tramontana presentation, special guest speech, high turnout, attendance sheet signing, community questions and ICS demonstration

Table 7. Some of the most common questions raised by local community members and responses given by the project team for various stakeholder meetings in the first project instance

Name of attendee	Question	Local coordinator response
Vanthala Lavanya from Baski village (Araku Valley)	How do traditional cookstoves and the Tramontana ICS differ?	As highlighted in the presentation, unlike traditional cookstoves which produce a lot of toxic localised air pollution and are low in quality, Tramontana ICS are made using steel/ceramic and fully combust the

		firewood to generate more heat and significantly less smoke. It also saves time on fuel gathering for the family as there is an estimated 60% reduction in firewood requirement. A list of the advantages can be seen in the brochure provided and on the banners displayed around the venue. You can discuss further details with any of the local representatives present here following this meeting.
G. Chandramma from Y.N Pakalu village (Koyyuru)	How do we fire up these ICS?	The Tramontana ICS can be fired up in the same way as your current cookstoves. The advantage is that you do not need to drastically change the way you cook or use the stove. We will provide instructions verbally and in a brochure format with every ICS during distribution, as well as give physical demonstrations, to help you safely and effectively operate and maintain the ICS. Also, a contact number will be provided in case you are having any issues or require advice.
B. Kondamma from Pedalochali village (G. Madugula)	How much weight can be put on the Tramontana ICS for cooking?	We estimate you can put up to 15 kg of weight on the Tramontana ICS, which should be more than sufficient to support cooking (and other activities) for the entire family.
D. Saraswathi from Dokuluru village (Paderu)	If there are 10 members in a family, how will a single ICS be sufficient?	A single ICS may in fact be sufficient due to its increased thermal efficiency and higher cooking capacity. However, we will consider providing an additional stove if the amount of firewood you are currently using is significantly higher than the average family in the community. For this we would have to take further fuel weight measurements and other details for evaluation¹².

¹² See section 5.4 for details on accounting for multiple devices per household in emissions reduction calculations

Konudi Pushpalatha from Dabavalasa village (Hukumpeta)	How long are the Tramontana ICS under warranty?	The average lifespan of the Tramontana ICS is estimated to be 7 years. We will provide you with a contact number at the time of distribution which you can call if at any time you are experiencing problems with your ICS. We will provide maintenance, repair and replacement services due to any malfunction or damage to the ICS free of cost. Our team will also conduct at least annual household visits to ensure you are using the ICS and are satisfied with it.
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2.1.3 Free Prior and Informed Consent

Obtaining consent	During stakeholder meetings, the benefits of the ICS devices are described and safe usage of the ICS is demonstrated. The meetings also represent an opportunity for stakeholders to raise any concerns or ask questions, as discussed in Section 2.1.2. These measures ensure that stakeholders are well informed, able to relay their views, and project activities are fully transparent. During distribution of the ICS, upon voluntary agreement to participation, an agreement is signed between the Project Proponent and each beneficiary. The terms of the agreement are outlined and discussed in detail prior to signing.
Outcome of FPIC	All beneficiaries voluntarily provided consent to their involvement in the project and the terms of the license agreement. The feedback was overwhelmingly positive for project participation based on the stakeholder meetings and pre-baseline surveys conducted. The project has not encroached on land, relocated people without consent, or forced physical or economic displacement. No conflicts have been recorded under the project to date.

2.1.4 Grievance Redress Procedure

Development process	The grievance redress procedure was developed based on consultations with stakeholders during the planning stages of the
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	project. The process ensures that any grievances raised by stakeholders are respected, recorded and addressed.
Grievance redress procedure	Any concerns or questions stakeholders may have following stakeholder meetings and during the project life, can be directly reported to the local mandal representative or communicated through a dedicated phone number. All points raised will be added by the team to a local grievance register. The information in the register is reviewed on a regular basis by the Project Proponent, with all concerns addressed and incorporated into the project subject to merit.

2.1.5 Public Comments

Comments received	Actions taken
None	No public comments were received during the public comment period which was from 16-June-2023 to 17-July-2023

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The Project Proponent has extensive expertise in risk management in financial markets, project development and the management of large teams. The local subsidiary CropZone, has strong relationships with the local communities and some experience in on-the-ground logistics and project implementation.

The Project Proponent is the coordinator and manager of the entire project. It makes all major decisions required in the project, and provides continuous guidance and support to the local teams.

The Project Proponent provides extensive training and support to all project teams, which is discussed in detail in section 5. The more senior mandal coordinators employed are given specific management training for managing the local field teams and tackling any issues when engaging with beneficiaries.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	The project directly improves the wellbeing of stakeholders through the provision of a Tramontana ICS. The ICS device is associated with cleaner combustion characteristics and increased thermal efficiency. This improves conditions in the home, as the amount of indoor air pollution is significantly reduced, and more family time is available owing to time savings from when collecting firewood and using the stove. The compact nature and improved design of the ICS also enhances safety and reduces burns/injuries
Risks to stakeholder participation	Risk identified	Multiple preventative measures have been taken to minimise risks associated with stakeholder participation. During stakeholder meetings, the benefits of using an ICS device and following sustainable practices are clearly communicated to families, alongside outlining the planned project activities. This ensures that the community members understand the goals of the project and are educated on the direct benefits to them. Throughout the project, regular monitoring ensures that the project status is continuously assessed, including tracking the number of ICS that are operational following installation. This ensures that any ICS found not to be in use are quickly identified and removed from emissions reduction calculations, and direct communication is made with the corresponding beneficiary to understand the reasons for not

		using their ICS. Employment of local project staff facilitates this process and ensures that any challenges in regards to the operation of an ICS (such as breakdown of the ICS, severe damage etc) are swiftly resolved.
Working conditions	No risk identified	The Tramontana ICS project does not attempt to change the working customs of local communities, including the general stove type or method of use. This ensures confidence and comfort to households when using their ICS. The ICS device is also associated with cleaner combustion characteristics and increased thermal efficiency. This improves working conditions in the home, decreasing indoor air pollution, increasing time availability and improving stove use safety.
Safety of women and girls	No risk identified	The project directly improves the safety and well-being of women and girls, who are the ones mostly undertaking stove-related tasks, involving usage, fuel collection and cleaning. The project reduces drudgery and exposure to risks including wild animal attacks, injuries from tree cutting/wood chopping etc. which have been observed from baseline surveys (see section 3.4).
Safety of minority and marginalized groups, including children	Risk reduced from baseline	During stakeholder meetings, explanations and demonstrations of the ICS are given to ensure that families are clear on the correct, safe operation to minimise risk of injury. Any clarity needed or difficulties faced during operation can also be discussed with a local representative or using the telephone number provided.

		The ICS device has generally been designed to be compact and enclose the flame (i.e. no open combustion) to reduce risk of burn or injury to users.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	The Tramontana ICS substantially reduces toxic indoor air pollution through its improved design and cleaner combustion characteristics. This is clearly evident from the reduction in smoke experienced and reduced staining of utensils/walls. Thus there are immediate benefits for project families, including reducing risk of respiratory and retinal diseases.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ¹³	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	As part of project activities, training programs are implemented for all field staff and coordinators employed. The training enforces strict rules regarding discrimination. Should any accusations be made by beneficiary families or other villagers, the specific employed individuals will be evaluated and immediately removed from the programme if found to be at fault. Since the onset of the project, no project stakeholders have reported any form of discrimination.

¹³ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Sexual harassment	No risk identified	As part of the training programs strict rules are communicated regarding sexual harassment. Should any accusations be made by beneficiary families or other villagers, the specific employed individuals will be evaluated and immediately removed from the programme if found to be at fault. There have been no project stakeholders that have reported any form of sexual harassment since the project start.
Equal pay for equal work	No risk identified	All employees under the project have been paid fairly and in accordance with local customs and workers' rights. Pay is awarded exclusively and equally based on the work completed.
Gender equity in labor and work	No risk identified	Employment of local community members and improving their livelihoods and skillset is a central component of the project, particularly given the lack of opportunities available to the marginalised project communities. Where possible the project also looks to employ female team members to improve the gender representation that are in work, which is typically low in the region.
Forced labor	No risk identified	Forced labour is not allowed in the project. Work policies and procedures including in relation to forced labour are communicated to teams involved in the project as part of training.
Child labor	No risk identified	Child labour is forbidden in the project. Background checks are conducted on all staff and a strict 18+ age policy is enforced. Work policies and procedures including in relation to child labour are communicated to teams involved in project work during training.

Human trafficking	No risk identified	Human trafficking is forbidden in the project. Background checks are conducted on all staff. Work policies and procedures including in relation to human trafficking are communicated to teams involved in project work during training.
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2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project recognises, respects, and promotes the protection of rights of all local communities and employees in line with applicable international human rights laws, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	All project activities are conducted following local customs/traditions, respecting the area's cultural heritage. Stakeholder meetings are also conducted following this and in the local language/dialect by team members employed specifically from the local community.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	All property rights are fully respected and are not impacted by the project activities

2.3.5 Benefit Sharing

This section is not applicable as the project does not impact property rights as discussed in Section 2.3.4.

2.4 Ecosystem Health

No negative environmental impacts were identified for the project. An Environmental Impact Assessment (EIA) is not required as there are only positive environmental outcomes.

Impact	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	No risk identified	Through the improved design and increased thermal efficiency of the ICS, there is a reduced firewood requirement. This results in reduced deforestation/forest degradation, leading to a positive impact on biodiversity and ecosystems.
Soil degradation and soil erosion	No risk identified	The reduced deforestation/forest degradation resulting from the lower firewood requirement, leads to less soil degradation and erosion.
Water consumption and stress	No risk identified	The better quality soil conditions as a result of the project activities lead to improved soil water retention. Less water is also needed for cleaning utensils and maintenance of the household owing to the cleaner combustion of the ICS and minimised smoke/soot generation. These points were also highlighted in the monitoring database by beneficiaries.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified	The project is not located in or adjacent to habitats for rare, threatened or endangered species

Areas for habitat connectivity	No risk identified	The project is not located in or adjacent to areas for habitat connectivity
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2.4.2 Introduction of Species

N/A

2.4.3 Ecosystem Conversion

N/A

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type	Reference ID	Title	Version
Methodology	VM0050	Energy Efficiency and Fuel-Switch Measures in Cookstoves ¹⁴	1.0
Methodology	CDM AMS-II G	Energy efficiency measures in thermal applications of non-renewable biomass ¹⁵	13.1
VCS Tool	VT0008	Additionality Assessment ¹⁶	1.0
CDM Tool	CDM TOOL30	Calculation of the fraction of non-renewable biomass	4.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VM0050	The project activity corresponds to: a) Replacement of non-renewable biomass (e.g. firewood, charcoal)-fired cookstoves with any of the following: (i) More efficient project devices that use the same fuel as the baseline;	The project activity involves the replacement of thermally inefficient wood-fired stoves with more efficient ICS, using the same fuel as the baseline. The project activity therefore meets a)(i) of the applicability condition.

¹⁴ <https://verra.org/wp-content/uploads/2024/10/VM0050-EE-and-Fuel-Switch-Measures-in-Cookstoves-v1.0.pdf>

¹⁵ <https://cdm.unfccc.int/UserManagement/FileStorage/6TOUCX21D0BHNVRZFWMEKALY94GS7>

¹⁶ <https://verra.org/wp-content/uploads/2024/10/VT0008-Additionality-Assessment-v1.0.pdf>

	(ii) Efficient project devices fired by renewable biomass or bioethanol; (iii) Efficient project devices fired by liquefied petroleum gas (LPG); or (iv) Electric-powered project devices b) Replacement of solid or liquid fossil fuel (e.g. coal, kerosene)-fired cookstoves with any of the following: (i) Efficient project devices fired by renewable biomass or bioethanol; (ii) Efficient project devices fired by LPG; or (iii) Electric-powered project devices	
VM0050	Project devices are used in households, communities, institutions or small or medium enterprises (SMEs), collectively referred to in this methodology as the “target population”	The project activity is implemented in rural households
VM0050	Project devices using renewable biomass (fuel-switch) or non-renewable biomass (improved efficiency) are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 25%	Tramontana ICS installed under the project activity are high-quality single-pot portable cookstoves, which use woody biomass as a fuel input in line with manufacturer specifications. The project ICS CRS-23 model has an efficiency of 36.01% obtained through independent verification via laboratory testing conducted

		at the Department of Mechanical Engineering at Jorhat Engineering College (see certificate in Appendix 2). A newer ICS STAR-25 model has been developed with an efficiency of 52.50% (see certificate in Appendix 3). The testing of STAR-25 was conducted independently by the School of Energy Science and Engineering at the Indian Institute of Technology.
VM0050	The project proponent designs incentive mechanisms to reduce the use of inefficient baseline devices and practices that can be replaced by the project devices and describes these mechanisms in the project description	During distribution a signed end user agreement includes provisions to (voluntarily agree to) remove or demolish pre-project stoves from households eligible for a new ICS. Each beneficiary is directly supported to understand and follow the terms. Local field employees regularly monitor the project households, and families are also provided with their phone number, for clarifications or in case of any issues faced with their ICS. This grievance mechanism ensures that any problems are swiftly resolved, to minimise any chance of reverting to baseline practices. Larger families (≥ 6) may also request an additional project device if, amongst other additional details, the

		amount of firewood used is measured to be significantly higher than the average across the local area. The provision of an additional device should help disincentivise the use of baseline devices.
VM0050	Where a project device has ended its technical life, the project proponent either replaces it with a comparable or better project device or retrofits its essential components to continue meeting the minimum service level requirements (i.e., thermal energy generation), otherwise no further emission reductions may be claimed for the project device	All project devices will be replaced either with a comparable or even better device at the end of their technical life. If the lifespan ends up being shorter, it will be repaired or replaced by the local field teams accordingly. The Project Proponent keeps an updated record of all repairs/replacements which can be reviewed by the VVB during verification.
VM0050	Project proponents implement a method for the distribution and identification of project devices that avoids double counting of emissions reductions by other mitigation actions and includes unique product identification on the stove itself at the time of distribution/sale (e.g.. program logo, alpha/numeric ID, and end-user location, such as geographical coordinates, complete address information)	Each ICS installed under the project is identifiable by the Project Proponent logo and a unique serial number which is displayed on the ICS itself as a QR code and in written form. This is linked to individual users using the Tramontana ICS software application which scans the QR code and records corresponding ICS beneficiary information including personal details, complete address, date of installation, geolocation of household, user/ID photos etc (see section 5.3.1 for full list). The data will be

		verifiable by the VVB through an online database of distribution records. These project design measures eliminate the risk of double counting. Further details on the ICS distribution process can be found in section 1.12.
VM0050	The project complies with any national, sub-national, or local regulations or guidance for the installation, commercialisation, distribution and use of improved cookstoves and/or fuel supply and use for the target population. National, regional, and local regulatory frameworks for the provision of the type of thermal energy services provided by the project activity must be documented	There exists no mandatory law or regulations requiring the introduction or use of ICS in India. The project is undertaken on a voluntary basis by the Project Proponent.
VM0050	Where project activities reduce emissions from non-renewable biomass, including firewood and charcoal, the risk of double counting is assessed on a national basis by evaluating at validation and crediting period renewal whether there are REDD+ projects or jurisdictional REDD+ programs whose boundary overlaps with the expected fuel source area of the project. The project proponent must report on the findings of this assessment for informational purposes in the project description	From review of Verra and other carbon registries there are no REDD+ or jurisdictional REDD+ projects located within the project region.
CDM AMS-II G	The aggregate energy savings of a single project activity shall not exceed the equivalent of 60 GWh per year or 180 GWh thermal per year in fuel input	The energy savings for a single project activity is found to be below 60 GWh per year. The calculations meeting such condition are given in the Emissions Reductions spreadsheet provided.

CDM AMS-II G	Non-renewable biomass has been used in the project region since 31st December 1989, using survey methods or referring to published literature, official reports or statistics	According to the UN Food and Agriculture Organisation, there was significant demand for firewood in India observed over the period 1970 to 1995 (Table 8¹⁷), and also a large percentage share of the energy consumption in rural households coming from firewood (Table 9¹⁸). The high demand has been a key driver of forests being cleared/degraded, as presented in the State of Forestry reports from Forest Survey of India¹⁹. Over the period 1993 to 1997, 671,000 hectares or equivalently over 1% of total forest cover (dense and open forests) was lost in India. In addition, 1,831,000 hectares of dense forests was either degraded or destroyed. In the first project instance location of Andhra Pradesh, the total forest cover loss over the same period was estimated to be 396,600 hectares. Therefore it is clear that biomass from non-renewable sources has been in use in the project region since at least as far back as 31st December 1989. For the period beyond 1997 data sources such as
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¹⁷ <https://www.fao.org/forestry/statistics/80570/en>

¹⁸ <https://wgbis.ces.iisc.ac.in/energy/HC270799/RWEDP/acrobat/fd49.pdf>

¹⁹ <https://fsi.nic.in/index.php>

		the Global Forest Watch²⁰ show that use of non-renewable biomass has increased. The Global Forest Watch report between 2001 to 2023 states that India lost 2.33Mha of tree cover, equivalent to a 6% decrease in tree cover since 2000.
CDM AMS-II G	For cases where the biomass is sourced from renewable sources, the project participants should use a corresponding Type I methodology	Not applicable as biomass is not sourced from renewable sources under the project activity
CDM AMS-II G	The VCS project document shall also explain how the proposed procedures prevent double counting of emission reductions, for example, to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices	A standard license agreement is signed physically (with written signature or thumbprint) by each ICS beneficiary which includes (amongst other conditions) provisions for ownership of all emissions reductions generated by the ICS to be solely with the Project Proponent. The license agreement is generated through the Tramontana ICS software application on a tablet which is then signed by the end user (see Figure 2). A signed pdf copy of all agreements is available in the distribution database. No other stakeholders or parties involved in the project have any rights to the carbon credits generated

²⁰ <https://www.globalforestwatch.org/dashboards/country/IND/?category=forest-change&location=WyJjb3VudHJ5IiwSU5EII0%3D>

		from the GHG emissions reductions.
VT0008	This tool is applicable under any of the following conditions: 1. The applicable methodology requires or permits the use of this tool; or 2. The VCS Program rules and requirements require or permit the use of this tool	VM0050 requires the use of this tool to assess the baseline scenario. Thus applicability condition 1) is met
CDM TOOL30	This tool may be used by: a) DNAs to submit region- or country-specific default fNRB values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or b) project participants to calculate project- or PoA-specific fNRB values.	The tool is used by the Project Proponent to calculate project specific fNRB values for each instance, as presented in section 5. Thus applicability condition b) is met.

Table 8. Total firewood demand in India over the period 1970 to 1995¹⁵

Year	1970	1975	1980	1985	1990	1995
Firewood Demand (million m ³)	154.99	173.36	194.42	214.49	237.63	260.77

Table 9. Estimate share of energy consumption of firewood in rural households¹⁶

Type	% share in 1978 -79	% share in 1992-93
Firewood logs	18.95	32.49
Firewood twigs	35.62	29.11

3.3 Project Boundary

The boundary for the grouped project can be represented by the geographical boundary of India as presented in Figure 13 and also outlined in section 1.13. In line with methodology requirements, a KML file has been submitted delimiting the geographical area of origin of the biomass fuel used in the project.

Each project instance will also have its own separate defined boundary within the overall boundary of the grouped project. For the first project instance this corresponds to the Paderu division of the Alluri Sitarama Raju district of Andhra Pradesh depicted in Figure 7 and represented by approximate coordinates of latitude 18.011959 and longitude 82.555030.



Figure 13. Delineation of geographical boundary (red) within which all activity under the grouped project is contained

The greenhouse gases to be included in the project boundary are presented in the table below and the corresponding impact of introducing the Tramontana ICS is illustrated in Figure 14.

Source		Gas	Included?	Justification/Explanation
Baseline	Thermal energy generation	CO ₂	Yes	Major source of emissions
		CH ₄	Yes	Major source of emissions
		N ₂ O	Yes	Major source of emissions
	Production of charcoal fuel	CO ₂	No	Not applicable. Wood is the only fuel used
		CH ₄	No	Not applicable. Wood is the only fuel used
		N ₂ O	No	Not applicable. Wood is the only fuel used
Project	Thermal energy generation	CO ₂	Yes	Major source of emissions
		CH ₄	Yes	Major source of emissions
		N ₂ O	Yes	Major source of emissions
	Transport of fuel (where applicable)	CO ₂	No	Not applicable. Wood sourced from nearby forests is the only fuel used
		CH ₄	No	Not applicable. Wood sourced from nearby forests is the only fuel used
		N ₂ O	No	Not applicable. Wood sourced from nearby forests is the only fuel used
	Production of fuel (where applicable)	CO ₂	No	Not applicable. Wood sourced from nearby forests is the only fuel used
		CH ₄	No	Not applicable. Wood sourced from nearby forests is the only fuel used
		N ₂ O	No	Not applicable. Wood sourced from nearby forests is the only fuel used
	Self-generated electricity (non-renewable energy for backup)	CO ₂	No	Not applicable. Project introduces biomass-fired improved cookstoves
		CH ₄	No	Not applicable. Project introduces biomass-fired improved cookstoves
		N ₂ O	No	Not applicable. Project introduces biomass-fired improved cookstoves
		CO ₂	No	Not applicable. Project introduces biomass-fired improved cookstoves

Source	Gas	Included?	Justification/Explanation
Grid electricity generation and distribution	CH ₄	No	Not applicable. Project introduces biomass-fired improved cookstoves
	N ₂ O	No	Not applicable. Project introduces biomass-fired improved cookstoves

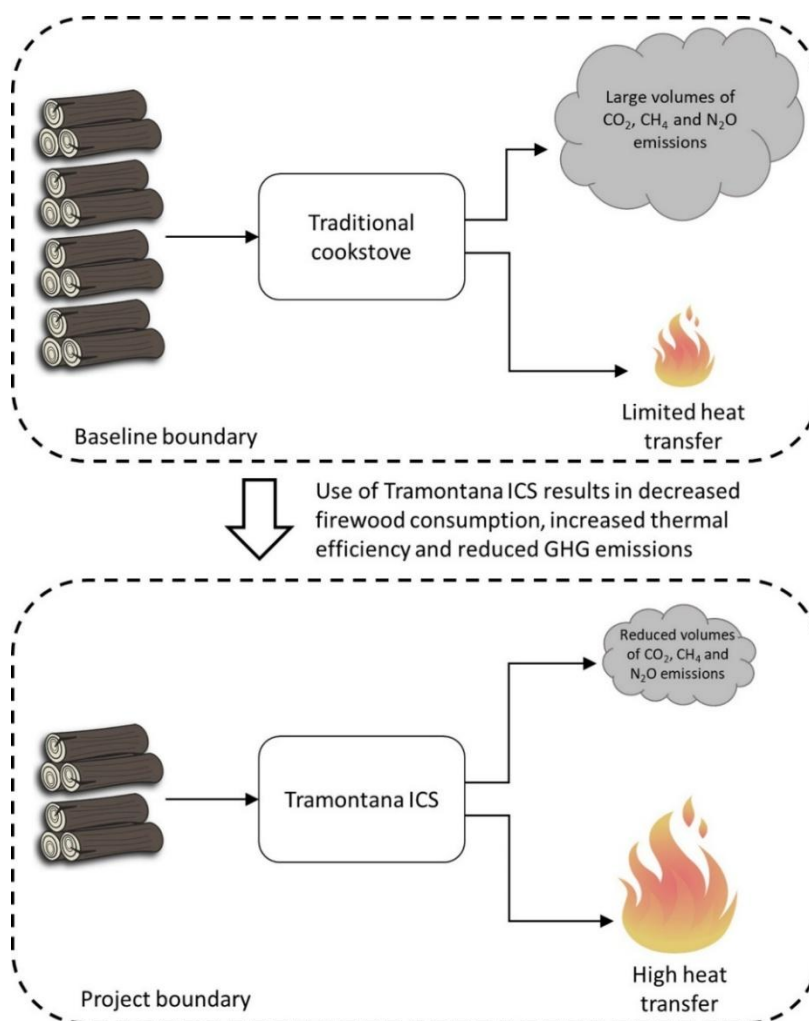


Figure 14. Depiction of the impact of introducing the Tramontana ICS on GHG emissions, firewood consumption and thermal efficiency in the project region

3.4 Baseline Scenario

3.4.1 Selection and justification of the baseline scenario

In line with VM0050, the selection and justification of the baseline scenario involves a three step process as outlined below. Alternative baseline scenarios must firstly be identified before being eliminated until only one feasible baseline scenario remains.

Step 1: Identify alternative baseline scenarios

- Alternative 1: The proposed project activity is undertaken without being registered under a GHG program
- Alternative 2: An alternative technology/fuel, such as LPG-fuelled stoves are introduced
- Alternative 3: The continuation of the current situation involving continued use of woody biomass as a fuel

Step 2: Consider existing and forthcoming government policies and legal requirements

As stated in section 1.15, there exists no mandatory laws or regulations requiring the introduction or use of ICS (including with alternative fuels) in India or for the necessary improvement of indoor air quality standards from stove usage. Therefore, none of the alternative baseline scenarios are eliminated at this step.

Step 3: Assess financial, institutional and information barriers

The barriers described below are realistic barriers described in section 6.1 of VM0050 and section 5.3.1 of VT0008 that may prevent the alternative scenarios above from occurring.

1. Financial barriers, for example:

- a. similar activities have only been implemented with grants or other non-commercial financing terms
- b. no capital is available from domestic or international capital markets due to real or perceived risks associated with investments in the applicable geographic area where the project activity is implemented

Alternative 1 faces significant financial barriers. The proposed project activity distributes ICS to communities free of charge and provides regular maintenance services throughout the project lifetime at no cost to the end user. Providing this service free of charge without the revenue generated from the sale of GHG carbon credits would not be financially feasible. Alternative 1 is therefore eliminated.

2. Institutional barriers, for example:

- a. the investor is not the beneficiary of financial or non-financial benefits associated with implementation of the project activity

- b. decentralised corporate structures that treat energy costs as an overhead instead of direct costs provide little incentive for organisational units to reduce energy use

Alternative 2 faces institutional barriers. Although LPG stoves offer an alternative solution to traditional cookstoves, due to the remote locations of the project communities sourcing the fuel is very difficult. Furthermore, as explained in 1.12, despite recent government efforts to promote cleaner cooking fuels (through schemes such as PMUY), challenges in uptake occur due to high fuel costs (i.e. for LPG refills). Families are unable to afford these fuels and also fail to understand the non-financial benefits. Firewood is also widely accessible and available free of cost, thus users have little motivation to purchase such alternative fuels.

3. Information barriers, for example:

- a. Lack of awareness of financial and non-financial benefits for final users
- b. Low acceptance of new or alternative practices, services or products associated with the project activity in the relevant sector due to lack of knowledge

Alternative 2 in addition faces information barriers. Families are often not willing to switch from their traditional practices to completely new unfamiliar technologies such as gas, for instance, due to cultural habits and preferences (e.g. firewood adding flavour to their food). Consequently Alternative 2 is eliminated.

The remaining baseline scenario is Alternative 3. This is the continued use of woody biomass by local communities to meet their thermal energy needs such as for cooking, heating and boiling water. This is the feasible baseline scenario and has been commonplace in the project areas for generations. Typically families use rudimentary traditional cookstoves in the form of fixed mud clay stoves (Figure 1 and Figure 15) or three-stone stoves (Figure 3) which use firewood as a fuel.

3.4.2 Baseline scenario survey

Despite all project activity instances sharing some comparable characteristics, separate baseline surveys are conducted for each instance in order to gauge an accurate representation of the current practices in that specific region. A similar approach to the baseline surveys conducted for the first project instance outlined below will be followed for other instances.

For the first project instance, baselines surveys were conducted across 11 mandals in the Alluri Sitarama Raju district of Andhra Pradesh. This occurred prior to distribution between 3rd November 2022 and 12th December 2022. As a first step, pre-baseline surveys were conducted to estimate the approximate number of potential ICS beneficiaries in each village. By applying random sampling to the pre-baseline data, 238 villages were randomly selected for baseline assessment and within each village an average of 5-10 surveys were conducted. This resulted in significant oversampling, with 1944 household surveys conducted which represented approximately 4% of all households for the instance. This is equivalent to 54 times the minimum sample size required to meet a 90/10 confidence/precision. The sample size calculation and statistical analysis methods used are presented in section 5.3.2.

To understand the baseline situation across the project instance area, detailed face-to-face surveys were conducted by trained local field teams employed for the project, using tablets containing the Tramontana ICS software application. All baseline records were signed, submitted and successfully synced to a Google cloud database before being automatically translated into an Excel sheet. The sheet was then checked by an office team to identify any incorrect entries and immediately have them resurveyed. The final baseline sheet is submitted to the VVB and Verra for the purposes of validation.

The baseline survey has been designed, carried out and analysed in line with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities and includes all questions as required from Appendix 3 of the VM0050 v1.0 methodology. Specific information collected includes (but is not limited to):

- Name and phone number of household representative
- Household size, composition, individual occupations and education levels
- Address of household (including GPS coordinates)
- Details on current fuel, including type, volumes used and time spent collecting
- Details on current stove including use cases, frequency of use and seasonality
- Photos of traditional stove and cooking area
- Perception of ICS and challenges of the traditional stove
- Date and time of interview
- Field team member information including name and title

From the baseline survey results for the first project instance, it is found that the vast majority of families (81%) are using traditional fixed mud clay stoves and a smaller proportion (19%) use three-stone stoves. This finding is supported by the 2019-2021 India National Family Health Survey²¹, which found that, of those using solid fuels across India, 99.3% were found to be using open fire or mud-clay stoves.

In our project region, we specifically selected firewood-using families (which are the vast majority), with LPG and other fuel users not included. Although some government reports suggest there is LPG usage in Andhra Pradesh, schemes such as PMUY, have been particularly unsuccessful in remote areas due to the high cost of fuel and difficulty in accessing refills. A press release from the Ministry of Petroleum and Natural Gas²² reported that in Andhra Pradesh alone, over 58% of PMUY beneficiaries have taken four or less refills and therefore have either reverted to firewood stoves or are stove stacking with firewood. In relation to the fuel source location, state-specific data was not available, although it was tentatively reported that a majority of rural households in India collect

²¹ <https://www.dhsprogram.com/pubs/pdf/FR375/FR375.pdf>

²² <https://www.pib.gov.in/PressReleasePage.aspx?PRID=1843444>

firewood (rather than purchasing it or receiving it as gifts etc)²³. In the project region, we found that in all cases the collection of wood was reported to be sourced directly from forests. This is commonplace across the region primarily owing to the communities being located in or close to forests.

From baseline surveys, it is found that over 86% of respondents are using their stove inside the household and for an average of 4.4 hours per day all year round. All respondents were reported to use their stove for the purposes of cooking, boiling water and in many cases (81%) also heating. Note that in 10% of the cases individuals also used their stoves to prepare food/beverages for sale. In addition to the above, an average of 14 hours per week is spent sourcing, carrying and processing (e.g. cutting, shredding) the firewood, which highlights the overall time-consuming nature of stove preparation tasks. It is exclusively women who were using the stoves (aside from a few cases where no women lived in the household), with the firewood gathering also undertaken by them, sometimes with the help of other family members. Generally, it was clear that stove-related activities formed the primary role of female members in the community.

Over 98% of respondents stated that they were not happy with their current stove and accordingly very few stated that it was not a priority to change. All participants confirmed that they did not own an ICS and would be interested to receive a new ICS if offered one.

A number of key issues were widely reported across the baseline surveys. These included: experiencing wild animal attacks in the forest (including snakes, bears, boars and dangerous insects), injuries when cutting wood using tools/axes, sexual assault, allergies from exotics plant species, toxic localised smoke causing retinal and breathing issues, and physical pain in undertaking the physically demanding tasks of cutting trees, carrying firewood long distances in hilly areas (average of 3.31km away) and sitting in a fixed position adjacent to the stove for prolonged periods.

Almost all respondents (93%) stated that there is a lot of localised smoke produced by their traditional stoves, 84% that the associated tasks are time consuming giving little time for other activities, and 62% that a large quantity of firewood is required to be regularly collected/used. Despite all respondents knowing that kitchen smoke has a negative impact on health, they have little ability to control it. When questioned on why they use traditional stoves which have such issues, the main reasons given were that they are not aware of any other stoves, alternative fuel sources are too expensive, firewood is abundantly available and/or that their current stove has been at least manageable in terms of handling/maintenance. Visual observations indicated that traditional stoves were producing large volumes of soot, often covering pots, utensils and walls in a thick black coating (see Figure 15).

All baseline respondents confirmed having a Below Poverty Line/ration card, and over 99% confirmed availability in physical form at the time of surveying. Where available these cards are photographed at the time of ICS distribution. Finally, over 99% of the respondents stated that their family's main work is in agriculture and the total income per family was found to average out to under one US dollar a day.

²³ https://link.springer.com/content/pdf/10.1007/978-3-031-71605-8_5.pdf?pdf=inline%20link

Following the release of the VM0050 methodology in October 2024, 151 Kitchen Performance Tests (KPTs) were conducted and 151 additional baseline surveys were carried out between 16th November to 13th December 2024 and 14th to 22nd June 2025. Note that the additional baseline surveys included two further questions required by the updated methodology:

1. How many meals did you prepare yesterday? Is this typical?
2. Is there any difference with the amount of meals prepared during a typical weekend day?

The specific KPT protocol followed is outlined in section 5.3. From the tests, it was found that an average of 13.22kg of woody biomass was being used per household per day in the baseline scenario of Instance 1. This equates to 4.83 tonnes per household per year.

Clarification 1 of *Correction and Clarifications to VM0050*²⁷ states that “baseline fuel consumption results must be compared with data from reputable sources, including government publications, peer-reviewed literature, third-party studies or official data and statistics”. This data must also have been “collected no more than five years ago”. While national-level data has been updated through various reports and surveys, state-level breakdowns post-2013 are scarce. The India State of Forest Report and National Sample Survey Office (NSSO) are the most reliable sources, but their latest publicly available datasets on fuelwood consumption remain based on earlier assessments. Consequently, the consumption results have instead been compared to an external study conducted in the project area between December 2024 and January 2025 by an independent specialist local company - Koshish Sustainable Solutions. The assessments were undertaken by Dr. Amit Ranjan Verma (PhD, IIT Delhi), supported by Dr. Ratnesh Tiwari (PhD, IIT Delhi) and Vishnu Chatriya (BA, Rajendra College Chapra), who as a team are renowned experts in biomass cookstove testing in both laboratory and in-situ environments. Their study found an average of 14.5kg of wood per household per day being used in the baseline scenario, which validated our field measurements (see section 5.3.2 for more detail and the full independent test report provided).

Of the 151 households in Instance 1 selected for baseline KPTs, specifically 51 remained outside of the project (i.e. did not receive a Tramontana ICS). These households, known as “control households”, are used to monitor any changes in stove usage characteristics in the Instance 1 area. If any changes are observed over time, for example, reduced fuel consumption or the adoption of an additional stove, the baseline fuel consumption for Instance 1 ($BC_{b,i,y}$) will be updated.

Of the KPT households surveyed, the average household size (Hh_i) was found to be 3.38 equivalent male adults, or 4 in absolute numbers. The 2020-2021 Multiple Indicator Survey in India²⁴ reported that the average household size in rural areas of Andhra Pradesh is 3.5, therefore supporting our observations.

²⁴ https://mospi.gov.in/sites/default/files/publication_reports/MultipleIndicatorSurveyinIndia.pdf



Figure 15. Baseline scenario depicting a traditional mud clay stove, firewood being transported by headload, cooking using a traditional stove and Tramontana meeting with ICS beneficiaries from a local indigenous community

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☒ No

3.5.2 Additionality Methods

The methodology uses the Activity Method to demonstrate additionality. This involves:

Step 1: Regulatory Surplus

The project is designed and implemented on a voluntary basis by the Project Proponent and therefore it is a voluntarily coordinated action.

There is no mandated government program or policy in the host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is also not mandated by any law, statute, or other regulatory frameworks, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or other regulatory framework.

Step 2: Positive List

The positive list represents the applicability conditions (outlined below) from the methodology which must be satisfied for a project to be deemed additional.

1. The project activity introduces:
 - a. Efficient biomass-fired cookstoves that replace inefficient biomass-fired cookstoves;
 - b. Efficient, solely renewable biomass-fired cookstoves that replace fossil fuel-fired cookstoves; or
 - c. Electric cookstoves that replace inefficient biomass-fired or fossil fuel-fired cookstoves
2. The project activity installs or distributes improved cookstoves at zero cost to the end user and has no revenue source other than from the sale of verified carbon units (VCUs).

The project introduces efficient biomass-fired cookstoves that replace inefficient biomass-fired cookstoves. The project ICS (and any other ICS that may be included in future instances) will be distributed and licenced to beneficiaries free of charge under the project activity, with no sources of revenue generated from the project activity other than from the sale of GHG carbon credits. As the project activity satisfies applicability conditions 1 and 2 above, it can therefore be deemed as additional.

Step 3: Project Method

Not applicable as the project passes Step 1 and Step 2 and is therefore deemed additional.

3.6 Methodology Deviations

The project activity has not taken any deviations from the applied methodology VM0050 v1.0.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

The following section describes how the emissions reductions for the project are calculated. Baseline and project emissions are first calculated separately in sections 4.1 and 4.2, and then combined to quantify total emissions reductions in section 4.4. Note that the following calculations are for ex-ante purposes only, with the actual emissions reductions achieved over each monitoring period to be presented in monitoring reports.

4.1 Baseline Emissions

Baseline emissions are calculated using Equation 1 from the VM0050 methodology:

$$BE_y = \sum_{i,j,k} EC_{i,y} \times N_{j,k,y} \times n_{j,k,y} \times (EF_{b,i,CO_2} \times f_{NRB,y} + EF_{b,i,nonCO_2}) \quad \text{Equation 1}$$

Where:

BE_y	=	Baseline emissions during year y (t CO ₂ e)
$EC_{i,y}$	=	Average energy consumption of baseline device type i in year y (TJ)
$N_{j,k,y}$	=	Number of commissioned project devices of type j from batch k in year y
$n_{j,k,y}$	=	Proportion of commissioned project devices of type j from batch k that remain operating in year y (fraction)
EF_{b,i,CO_2}	=	CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO ₂ /TJ)
$f_{NRB,y}$	=	Fraction of woody biomass that is established to be non-renewable used by baseline device in year y (fraction)
$EF_{b,i,nonCO_2}$	=	Non-CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO ₂ e/TJ)
i	=	Baseline device type and its respective fuel type
j	=	Project device type and its respective fuel type

The average energy consumption of baseline devices is determined using:

$$EC_{i,y} = BC_{b,i,y} \times NCV_{b,i} \quad \text{Equation 2}$$

Where:

$BC_{b,i,y}$ = Fuel used per baseline device type i during year y (tonnes)

$NCV_{b,i}$ = Net calorific value of baseline fuel for baseline device type i (TJ/tonne)

Parameter	Value	Source
$BC_{b,i,y}$	4.83 tonnes	Baseline KPT Survey – Instance 1
$NCV_{b,i}$	0.0156 TJ/tonne	Methodology

$$EC_{i,0} = 4.83 \times 0.0156$$

$$= 0.0753 \text{ TJ}$$

$EC_{i,y}$ must then be cross-checked to identify if there is any potential stove stacking. This is done by comparing the quantity of baseline energy consumption to that from the back-calculation of the project scenario based on stove thermal efficiencies. If results indicate a baseline consumption that is higher than that indicated through the back-calculation, the latter is applied as a conservative cap. The back-calculation is given by:

$$EC_{est,y} = EC_{p,y} \times \frac{\eta_{new,avg,y}}{\eta_{old,avg}} \quad \text{Equation 3}$$

Where:

$EC_{est,y}$ = Back-calculated energy consumption of the potential mix of devices and fuels in the baseline in year y (TJ)

$EC_{p,y}$ = Energy used in project scenario by project devices during year y (TJ)

$\eta_{new,avg,y}$ = Weighted average efficiency of project devices in year y (fraction)

$\eta_{old,avg}$ = Weighted average efficiency of baseline devices that are replaced by project devices (fraction)

$EC_{p,y}$ is determined using:

$$EC_{p,y} = \sum_{j,k} BC_{p,j,k,y} \times NCV_{p,j} \quad \text{Equation 4}$$

Where:

$BC_{p,j,k,y}$ = Average quantity of fuel used by project device type j from batch k during year y (tonnes or m³)

$NCV_{p,j}$ = Net calorific value of project fuel used in project device type j (TJ/tonne or TJ/m³)

As an example, for Instance 1 at year $y = 0$, we use the following parameters and apply the calculation to determine if the cap is applied:

Parameter	Value	Source
$BC_{p,j,k,y}$	1.75 tonnes	Project KPT Survey – Instance 1
$NCV_{p,j}$	0.0156 TJ/tonne	Methodology
$\eta_{new,avg,0}$	36.01%	Water Boiling Test for the CRS-23 stove (see Appendix 2)
$\eta_{old,avg}$	15%	Methodology

$$EC_{p,0} = 1.75 \times 0.0156$$

$$= 0.0273 \text{ TJ}$$

$$EC_{est,0} = 0.0273 \times \frac{0.3601}{0.15}$$

$$= 0.0656 \text{ TJ}$$

As $EC_{est,0} < EC_{i,0}$, a cap is applied and $EC_{i,0}$ is taken as 0.0656 TJ

To then calculate BE_y in Equation 1, $EC_{i,y}$ here is used along with the other parameter values given in Table 10 later in this section. The calculation is presented in section 4.4.

4.2 Project Emissions

Project emissions, corresponding to the emissions generated during use of project ICS devices, are calculated using Equation 6 from the methodology:

$$PE_y = PE_{energy,y} + PE_{others,y} \quad \text{Equation 5}$$

Where:

PE_y	=	Project emissions during year y (t CO ₂ e)
$PE_{energy,y}$	=	Project emissions from energy consumption of project devices in year y (t CO ₂ e)
$PE_{others,y}$	=	Project emissions from other sources in year y (t CO ₂ e)

As the project ICS devices use biomass fuel, the component of project emissions from energy consumption is determined using:

$$PE_{energy,y} = \sum_j \sum_k BC_{p,j,k,y} \times N_{j,k,y} \times NCV_{p,j} \times n_{j,k,y} \times (EF_{p,j,CO_2} \times f_{NRB,y} + EF_{p,j,nonCO_2}) \quad \text{Equation 6}$$

Where:

$BC_{p,j,k,y}$	=	Average quantity of fuel used by project device type j from batch k during year y (tonnes or m ³)
$N_{j,k,y}$	=	Number of commissioned project devices of type j from batch k in year y
$NCV_{p,j}$	=	Net calorific value of project fuel used in project device type j (TJ/tonne or TJ/m ^{3A})
$n_{j,k,y}$	=	Proportion of commissioned project devices of type j from batch k that remain operating in year y (fraction)
EF_{p,j,CO_2}	=	CO ₂ emission factor for fuel used by project device type j in the project scenario (t CO ₂ /TJ)
$f_{NRB,y}$	=	Fraction of woody biomass that is established to be non-renewable used by device in year y (fraction)
$EF_{p,j,nonCO_2}$	=	Non-CO ₂ emission factor for fuel used by project device type j (t CO ₂ /TJ)

$PE_{others,y}$ is calculated using the following Equation:

$$PE_{others,y} = PE_{transp,y} + PE_{prod,y} + PE_{fugitive,y} + PE_{backup,y} \quad \text{Equation 7}$$

Where:

$PE_{transp,y}$	=	Project emissions due to fuel transportation in year y (t CO ₂ e)
$PE_{prod,y}$	=	Project emissions due to fuel production in year y (t CO ₂ e)
$PE_{fugitive,y}$	=	Fugitive emissions in year y (t CO ₂ e)
$PE_{backup,y}$	=	Project emissions due from backup generators in year y (t CO ₂ e)

As the project devices use biomass collected from local forest areas typically by headload, the value of $PE_{others,y}$ is zero. Therefore, $PE_y = PE_{energy,y}$.

The calculation of PE_y has been detailed alongside the baseline emissions in section 4.4.

4.3 Leakage Emissions

Leakage is accounted for using the adjustment factor of 0.95 in line with the methodology. This value is considered conservative, as there is little incentive for non-project users to switch fuels or increase consumption given that woody biomass is already widely available at zero cost across the project region. However, the project developer will continually assess all potential leakage sources and disclose any related findings in monitoring reports.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

In line with VM00050, GHG emissions reductions are calculated using the following equation:

$$ER_y = (BE_y - PE_y) \times 0.95 - LE_{RB,y} \quad \text{Equation 8}$$

Where:

ER_y	=	Emission reductions during year y (t CO ₂ e)
$LE_{RB,y}$	=	Leakage emissions associated with use of renewable biomass during year y (t CO ₂ e)

$LE_{RB,y}$ is zero as the project does not involve the use of renewable biomass.

The following assumptions are applicable for the project activity to help estimate the emissions reductions ER_y for ex-ante purposes:

- Project activity installs up to 400,000 ICS across the project region

- Commencement of ICS installation for a project activity instance is considered as the operational date of that project instance, and the date of installation of the last ICS within an instance determines the expected operational lifetime of the instance
- An expected average lifespan for an ICS of 7 years. This can be extended with regular maintenance, and information on any lifetime extensions will be reported during monitoring.
- The baseline and project stove use the same fuel which is wood
- An annual stove loss rate of $L = 0$, as the project provides maintenance of all ICS and repair/replacement in case of any damage or stove failure. Any issues found by an ICS user can be communicated in person or through a dedicated contact number provided. A physical check is also undertaken with every household at least once a year. Note that the parameter L exists for ex-ante calculation purposes only and is not directly determined for the project. Instead it is incorporated within the parameter $N_{j,k,y}$ which is monitored through the project life.
- $BC_{ex-ante,b,i}$ is calculated using a KPT conducted prior to ICS installation for each project instance. The value 4.83 tonnes per year output from results for the first instance (see section 3.4) is used here.
- A ramp-up schedule is defined for ICS distribution with 100,000 over the first year, followed by 200,000 in year 2 and 100,000 in year 3 reaching a total of 400,000 ICS.
- The first 150,000 ICS distributed will be CRS-23 and the subsequent 250,000 will be STAR-25 in line with the roll out schedule described in section 1.12
- For simplicity, ex-ante calculation workings in this section do not delineate project instances k

Calculation of f_{NRB}

The fraction of non-renewable biomass required to determine BE_y and PE_y has been calculated in line with the methodology using CDM TOOL30, with the application of an uncertainty discount of 26%.

The following calculations to determine f_{NRB} are provided in an Excel sheet separately.

$$f_{NRB} = \frac{NRB}{NRB + RB} \quad \text{Equation 9}$$

Where:

f_{NRB}	=	Fraction of non-renewable biomass in the applicable area in the relevant period (fraction or %)
NRB	=	Quantity of non-renewable biomass consumed in the applicable area in the relevant period (tonnes)
RB	=	Quantity of renewable biomass that is available on a sustainable basis in the applicable area in the relevant period (tonnes)

The quantity of non-renewable biomass (NRB) is determined using:

$$NRB = H - RB \quad \text{Equation 10}$$

Where:

H = Total consumption of woody biomass in the applicable area in the relevant period (tonnes)

The total consumption of woody biomass (H) is calculated through the following equation, accounting for all consumption within the applicable area (which includes timber and industrial consumption):

$$H = HW \times N + CE + NE \quad \text{Equation 11}$$

Where:

HW = Average consumption of wood fuel per household, including fuelwood and charcoal, in the applicable area in the relevant period (tonnes/household)

N = Number of households consuming wood fuel within the applicable area in the relevant period (number)

CE = Commercial woody biomass consumption for energy applications (e.g. commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

NE = Commercial woody biomass consumption for non-energy applications (e.g. construction, furniture) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

For Instance 1, the parameters used to calculate H are listed in the table below. Note that these parameters are area-specific and are updated in future monitoring reports for new instances.

Parameter	Value	Source	Reference
HW	4.83 tonnes /household	Baseline KPT Instance 1	Average pre-project fuel usage
N	4,093,769	Multiple - See f_{NRB} Excel sheet for exact calculation	-

<i>CE</i>	0	Data not available, so conservatively assuming zero	-
<i>NE</i>	16,412 tonnes	Calculated from Andhra Pradesh Forest Department data	See f _{NRB} Excel sheet for exact calculation

Based on the values from the table, we calculate:

$$H = 4.83 \times 4,093,769 + 0 + 16,412$$

$$H = 19,770,516 \text{ tonnes}$$

The quantity of renewable biomass (RB) from Equation 10 is determined through:

$$RB = \sum_i (MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})) + \sum_i (MAI_{other,i} \times (F_{other,i} - P_{other,i})) \quad \text{Equation 12}$$

Where:

$MAI_{forest,i}$ = Mean Annual Increment of woody biomass growth per hectare in subcategory *i* of forest areas in the relevant period (tonnes/ha/yr)

$MAI_{other,i}$ = Mean Annual Increment of woody biomass growth per hectare in subcategory *i* of other land areas in the relevant period (tonnes/ha/yr)

$F_{forest,i}$ = Extent of forest in sub-category *i* in the relevant period (ha)

$F_{other,i}$ = Extent of other land in sub-category *i* in the relevant period (ha)

$P_{forest,i}$ = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (in subcategory *i*) in the relevant period (ha)

$P_{other,i}$ = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other land areas (in subcategory *i*) in the relevant period (ha)

For Instance 1, the parameters used to calculate *RB* are given in the table below. These are area-specific and to be updated in future monitoring reports for new instances.

Parameter	Value	Source	Reference
$MAI_{forest,i}$	4.63 tonnes/ha/year	2006 IPCC Guidelines for National Greenhouse Gas Inventories ²⁵	See f_{NRB} Excel sheet for exact calculation
$MAI_{other,i}$	4.63 tonnes/ha/year	2006 IPCC Guidelines for National Greenhouse Gas Inventories ²⁵	See f_{NRB} Excel sheet for exact calculation
$F_{forest,i}$	3,725,800 ha	India State of Forest Report 2023 ²⁶	Table 1.3 (Total RFA Andhra Pradesh)
$F_{other,i}$	800,400 ha	India State of Forest Report 2023 ²⁶	Table 5.1 (2023 TOF - Andhra Pradesh)
$P_{forest,i}$	3,195,900 ha	India State of Forest Report 2023 ²⁶	Table 1.3 (RF - Andhra Pradesh)
$P_{other,i}$	0	Data not available, so conservatively assuming zero	-

Based on the values from the table, we calculate:

$$RB = (4.63 \times (3,725,800 - 3,195,900)) + (4.63 \times (800,400 - 0))$$

$$RB = 6,152,638 \text{ tonnes}$$

Therefore using Equation 10, NRB is determined as:

$$NRB = 19,770,516 - 6,152,638$$

$$NRB = 13,617,879 \text{ tonnes}$$

Applying the 26% uncertainty discount from the methodology in combination with Equation 9, f_{NRB} is calculated as:

²⁵ https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf

²⁶ https://fsi.nic.in/uploads/isfr2023/isfr_book_eng-vol-1_2023.pdf

$$f_{NRB} = \frac{13,617,879}{13,617,879 + 6,152,638} \times (1 - 0.26)$$

$$f_{NRB} = 0.510$$

Estimation of $N_{j,k,y}$

To calculate the ex-ante emissions reductions, next the expected number of ICS operating each year, $N_{j,k,y}$, must be estimated. For ex-ante calculation purposes, we use:

$$N_{j,k,y} = N_{j,k,0}(1 - y_l L) \quad \text{Equation 13}$$

Where:

$N_{j,k,y}$ = Number of commissioned project devices of type j from instance k in year y

L = Annual loss rate of stoves in the project area

y_l = Years of operation

It follows that:

$$n_{j,k,y} = \frac{N_{j,k,y}}{N_{j,k,0}} \quad \text{Equation 14}$$

Applying Equations 13 and 14 with the distribution schedule stated above with single stove type, for example for $y = 2$,

$$N_{1,2} = 100,000 \times (1 - 2 \times 0) + 200,000 \times (1 - 1 \times 0) = 300,000,$$

and assuming all ICS are utilised by users,

$$n_{1,2} = \frac{300,000}{300,000} = 1$$

However, in line with the *Correction and Clarifications to VM0050*²⁷ document and in section 5.3.2, $n_{j,k,y}$ is capped at 0.9. Therefore, the expected number of ICS operating at the end of each year is:

y	$N_{1,y}$	$n_{1,y}$
1	100,000	0.9
2	300,000	0.9
3	400,000	0.9

²⁷ https://verra.org/wp-content/uploads/2025/02/CC_VM0050_v1.0_Feb2025.pdf

4	400,000	0.9
5	400,000	0.9
6	400,000	0.9
7	400,000	0.9

Estimation of $BC_{p,j,k,y}$

To then estimate parameter $BC_{p,j,k,y}$ required to determine energy consumption $EC_{p,y}$ and therefore PE_y , the average efficiency of the ICS in year y , $\eta_{new,avg,y}$, from Equation 3 must be calculated. Throughout the project life, the average energy consumption of the project devices is determined via regular in-situ KPTs, as detailed in section 5. However, for the purposes of ex-ante calculations here, we determine energy consumption using the thermal efficiency and applying a linear annual reduction factor to the project ICS efficiency each year y .

Following the methodology outlined in paragraph 37 (a) of AMS-II G, the factor is obtained by taking the difference between the initial and final efficiencies, and dividing by the lifespan of the project device. Using the CRS-23 stove as an example: by taking the initial device efficiency of the project stove of 36.01% (from Table 2 and 10), a terminal efficiency of 25% as per AMS-II G guidelines, and a life span of 7 years, an annual efficiency reduction of 1.57% is calculated. Using this reduction factor, the efficiencies at the start ($\eta_{new}Start$) and end ($\eta_{new}End$) of each year have been determined, and an average ($\eta_{new,avg,y}$) taken to populate the values in the table below. The full calculation is provided in the ex-ante calculation sheet including the corresponding calculation for the STAR-25 stove.

y	$\eta_{new}Start$	$\eta_{new}End$	$\eta_{new,avg,y}$
1	36.01%	34.44%	35.22%
2	34.44%	32.86%	33.65%
3	32.86%	31.29%	32.08%
4	31.29%	29.72%	30.51%
5	29.72%	28.15%	28.93%
6	28.15%	26.57%	27.36%
7	26.57%	25.00%	25.79%

Based on these annual ICS efficiency estimates, the following equation is then applied to determine the mass of woody biomass used by an ICS in tonnes each year:

$$BC_{p,j,k,y} = BC_{ex-ante,b,i} \times \frac{\eta_{new,avg,y}}{\eta_{old,avg}} \quad \text{Equation 15}$$

Where:

$\eta_{old,avg}$ = Efficiency of baseline cookstove (%)

$\eta_{new,avg,y}$ = Efficiency of project stove in year y (%)

$BC_{ex-ante,b,i}$ = Ex-ante fuel consumption per baseline device type i (tonnes)

As an example, taking a value of $BC_{ex-ante,b,i} = 4.83$ tonnes (see section 5.1), $\eta_{old,avg} = 0.15$ in line with the methodology, and the value for $\eta_{new,avg,1}$ for $y = 1$ from the table above, we can calculate

$$BC_{p,j,k,1} = 4.83 \times \frac{0.15}{0.3522}$$

= 2.055 tonnes (3 d.p.)

Applying the formulation for each year, we get:

y	$BC_{ex-ante,b,i} \text{ (t)}$	$\eta_{new,avg,y}$	$BC_{p,j,k,y} \text{ (t)}$
1	4.83	0.3522	2.055
2	4.83	0.3365	2.151
3	4.83	0.3208	2.256
4	4.83	0.3051	2.373
5	4.83	0.2893	2.502
6	4.83	0.2736	2.646
7	4.83	0.2579	2.807

Final ER_y Calculation

Taking the values for f_{NRB} , $BC_{p,j,k,y}$, $N_{j,k,y}$ together with the fixed parameters outlined in Table 10 below, we can use Equation 8 to calculate the overall emissions reductions ER_y in year y .

Table 10. Fixed parameters and their source to calculate emissions reductions under the methodology

Parameter	Value	Source
$BC_{ex-ante,b,i}$	4.83 tonnes	Baseline KPT Instance 1
$\eta_{new,avg,0}$	36.01%	Manufacturer, Independent laboratory test
$\eta_{old,avg}$	15%	Methodology
$NCV_{b,i}$	0.0156 TJ/tonne	Methodology
$NCV_{p,j}$	0.0156 TJ/tonne	Methodology
EF_{b,i,CO_2}	112 tCO ₂ /TJ	Methodology
EF_{p,j,CO_2}	112 tCO ₂ /TJ	Methodology
$EF_{b,i,nonCO_2}$	9.46 tCO ₂ /TJ	Methodology
$EF_{p,j,nonCO_2}$	9.46 tCO ₂ /TJ	Methodology
f_{NRB}	0.510	Methodology Tool

As an example, for $y = 1$ we obtain an estimate for emissions reductions:

$$\begin{aligned}
 BE_{1,1} &= 4.83 \times 0.0156 \times 100,000 \times 0.9 \times (112 \times 0.510 + 9.46) \\
 &= 450,851 \text{ tCO}_2e
 \end{aligned}$$

$$\begin{aligned}
 PE_{1,1} &= 2.055 \times 0.0156 \times 100,000 \times 0.9 \times (112 \times 0.510 + 9.46) \\
 &= 191,995 \text{ tCO}_2e
 \end{aligned}$$

$$\begin{aligned}
 ER_{1,1} &= (450,851 - 191,995) \times 0.95 \times 0.5 \\
 &= 122,956 \text{ tCO}_2e
 \end{aligned}$$

An additional factor 0.5 in the equation above is applied given ICS distribution occurs throughout the year. The factor is to be applied for all ICS distribution years in line with the distribution ramp-up schedule.

Using the above approach, the ex-ante GHG emissions reductions for the proposed project activity (split by year of ICS distribution) over the first crediting period is given as:

y	N _{y,1}	ER ₁ (tCO ₂ e)	ER ₂ (tCO ₂ e)	ER ₃ (tCO ₂ e)	ER _y (tCO ₂ e)
1	100,000	122,956	-	-	122,956
2	300,000	237,387	287,362	-	524,749
3	400,000	228,026	554,387	150,589	933,002
4	400,000	217,699	530,673	290,462	1,038,834
5	400,000	206,250	502,617	277,773	986,640
6	400,000	193,484	468,835	262,512	924,830
7	400,000	179,161	427,258	243,806	850,225

The table below summarises the estimated baseline, project and emissions reductions over the first verification period separated by vintage. Detailed calculations are provided in the separate ex-ante calculations sheet.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
21-Jan-2023 to 31-Dec-2023	201,074	85,628	5,772	109,674	—	109,674
01-Jan-2024 to 31-Dec-2024	852,303	332,512	25,990	493,801	—	493,801
01-Jan-2025 to 31-Dec-2025	1,552,235	586,448	48,289	917,498	—	917,498

01-Jan-2026 to 31-Dec- 2026	1,802,708	707,070	54,782	1,040,856	—	1,040,856
01-Jan-2027 to 31-Dec- 2027	1,803,404	761,783	52,081	989,540	—	989,540
01-Jan-2028 to 31-Dec- 2028	1,803,404	826,284	48,856	928,264	—	928,264
01-Jan-2029 to 31-Dec- 2029	1,803,404	904,068	44,967	854,369	—	854,369
01-Jan-2030 to 20-Jan- 2030	100,189	50,468	2,486	47,235	—	47,235
Total	9,918,722	4,254,261	283,223	5,381,237	—	5,381,237

The estimated annual average GHG emissions reductions over the first crediting period is calculated to be 768,748 tCO₂e.

5 MONITORING

The following section outlines the monitoring framework for the project, first stating the parameters available at validation (section 5.1), next the parameters to be monitored at each verification stage (section 5.2) and finally the monitoring plan including data recording and sampling methods (section 5.3).

5.1 Data and Parameters Available at Validation

Data / Parameter	EF_{b,i,CO_2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for fuel used by baseline device type <i>i</i> in the baseline scenario
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas ²⁸
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	Wood is the fuel used in the baseline scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied
Purpose of data	Calculation of baseline emissions
Comments	--

Data / Parameter	EF_{p,j,CO_2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for fuel used by project device type <i>j</i> in the project scenario
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas ²⁸
Value applied	112

²⁸ https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/2_Volume2/19R_V2_2_Ch02_Stationary_Combustion.pdf

Justification of choice of data or description of measurement methods and procedures applied	Wood is the fuel used in the project scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied
Purpose of data	Calculation of project emissions
Comments	--

Data / Parameter	$EF_{b,i,nonCO2}$
Data unit	tCO ₂ e/TJ
Description	Non-CO ₂ emission factor for fuel used by baseline device type <i>i</i> in the baseline scenario
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas ²⁸
Value applied	9.46
Justification of choice of data or description of measurement methods and procedures applied	Wood is the fuel used in the baseline scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied
Purpose of data	Calculation of baseline emissions
Comments	--

Data / Parameter	$EF_{p,j,nonCO2}$
Data unit	tCO ₂ e/TJ
Description	Non-CO ₂ emission factor for fuel used by project device type <i>j</i> in the project scenario
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas ²⁸
Value applied	9.46
Justification of choice of data or description of	Wood is the fuel used in the project scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied

measurement methods and procedures applied	
Purpose of data	Calculation of project emissions
Comments	--

Data / Parameter	$NCV_{b,i}$
Data unit	TJ/tonne
Description	Net Calorific Value of baseline fuel used by baseline device type i
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas ²⁸
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	Wood is the fuel used in the baseline scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied
Purpose of data	Calculation of baseline emissions
Comments	--

Data / Parameter	$NCV_{p,j}$
Data unit	TJ/tonne
Description	Net Calorific Value of project fuel used by project device type j
Source of data	Option 3 as specified by the methodology guidelines: default value for wood fuel from the 2019 refinement of the 2006 IPCC Guidelines for National Greenhouse Gas ²⁸
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	Wood is the fuel used in the project scenario and therefore, in line with the methodology, the IPCC default value stated in the methodology has been applied
Purpose of data	Calculation of project emissions
Comments	--

Data / Parameter	$BC_{ex-ante,b,i}$
Data unit	tonnes
Description	Ex-ante annual average quantity of fuel used per baseline device type i
Source of data	Option 1 as specified in the VM0050 v1.0 methodology guidelines: Kitchen Performance Test (See “ICS-01 KPT Instance 1”)
Value applied	4.83
Justification of choice of data or description of measurement methods and procedures applied	The parameter value used is from the measured in-situ KPT results for 151 households sampled from the first project instance. The baseline KPT for Instance 1 was conducted between 13th November and 13th December 2024 and 11th to 16th June 2025. KPTs were exclusively conducted by local employees who received specialist training in line with the Kitchen Performance Protocol v4.0 published in March 2018. Based on the sampling calculation outlined in Appendix 5 of the methodology, a minimum sample size of 36 with 90/10 confidence precision was calculated (see section 5.3.2 for detailed calculations). KPTs were conducted for 151 households in Instance 1, therefore achieving an oversampling rate of 4.2. Detailed analysis has been conducted on the parameter which is presented in the excel sheet “ICS-01 KPT Instance 1” provided and in section 5.3.2 of this document. We observed a standard deviation 0.9894, confidence interval 0.265 and precision (2-sided) 3.35%. Therefore the 90/10 requirement of the methodology is achieved.
Purpose of data	Calculation of baseline emissions
Comments	Note that 51 of the KPTs conducted are in households which remain outside of the project (i.e. not receiving a Tramontana ICS) and act as “control households” in accordance with the methodology

Data / Parameter	$\eta_{old,avg}$
Data unit	Fraction

Description	Weighted average efficiency of baseline devices that are replaced by project devices
Source of data	Option 1 as specified in the VM0050 v1.0 methodology
Value applied	15%
Justification of choice of data or description of measurement methods and procedures applied	The default value for three-stone fires using firewood or cookstoves with no improved combustion air supply or flue gas ventilation is selected
Purpose of data	Calculation of baseline emissions
Comments	--

5.2 Data and Parameters Monitored

Data / Parameter	$N_{j,k,y}$
Data unit	Number
Description	Number of commissioned project devices of type j from batch k in year y
Source of data	Distribution database
Description of measurement methods and procedures to be applied	Distribution records are created through surveys conducted when each ICS is successfully distributed to a project beneficiary at their household This parameter is not calculated on a sample basis, but instead on the total project population. Thus it does not form part of sample-based monitoring in section 5.3.
Frequency of monitoring/recording	At the time of commissioning/distribution of ICS
Value applied	Sum of the entries in the distribution database over the period
Monitoring equipment	Distribution records are created using Samsung Galaxy A9+ tablets with the Tramontana ICS software application. Submitted records are stored locally on the tablet and automatically synced to a cloud database once the device is connected to the internet in the local offices. The automation of syncing maintains the data integrity and minimises any risk of data loss, theft, tampering or errors in translation. Further details on data reliability and traceability are explained in section 5.3.2

QA/QC procedures to be applied	During distribution an extensive dataset is obtained on each end user, including personal details, ID card/community certificate information and photo, and a photo of the individual together with their new ICS
Purpose of data	Calculation of baseline and project emissions
Calculation method	--
Comments	--

Data / Parameter	$n_{j,k,y}$
Data unit	Fraction
Description	Proportion of commissioned project devices of type j from batch k that are still being used regularly in year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	The parameter is monitored based on representative sampling of the database of distribution records. As per the methodology, the sampling complies with the latest version of CDM guidelines²⁹. Through the surveys, beneficiaries are questioned on their ICS usage practices over the monitoring period. If it is found that a non-project stove is being used following conducting monitoring surveys, the parameter $n_{j,k,y}$ is adjusted using a ratio of usage frequency of the project ICS compared to the frequency with which all stoves are used over a defined time period. For instance for a given household, if it is found from a survey that a non-project stove is being used 7 times per week and the ICS is being used 21 times a week, the following factor would be used as part of calculating $n_{j,k,y}$: $q = \frac{21}{21 + 7} = 0.75$ The values across all households monitored is then averaged to obtain a value for $q_{j,k,y}$
Frequency of monitoring/recording	At least annually
Value applied	Capped at 0.9 or lower if a sufficient number of pre-project or other devices are found to be in use based on results from

²⁹ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file_20151023152925068/Meth_GC48_%28ver04.0%29.pdf

	monitoring surveys. This is in line with the cap described in the <i>Correction and Clarifications to VM0050</i>²⁷ document published in February 2025. See section 5.3.2 for more details. Aiming for all ICS distributed to be operational owing to regular maintenance/repair and replacement if necessary.
Monitoring equipment	Trained field staff inspect the household to identify whether the ICS is in operation including taking photos of the stove, its components and the cooking area. A detailed survey using the Tramontana ICS software application is also conducted with the end user to identify current usage patterns and operation/performance over the past year.
QA/QC procedures to be applied	Access to the relevant experienced local staff member and their phone number is provided to every end user in case of any problems faced. This ensures that the Project Proponent can be directly contacted in order to resolve any issues promptly and make sure stoves remain in operation. Before conducting any monitoring surveys, field staff are also given specific training on the process and purpose of visits. This facilitates standardisation and a high quality being maintained. In addition to the required mandatory monitoring under the methodology, local staff complete additional internal checks with the aim of covering the entire population annually. Note that in this process, the sampled monitoring households are also revisited to verify the accuracy of existing data collected. More generally, the project aims to exceed the required sample confidence thresholds by using techniques such as oversampling.
Purpose of data	Calculation of baseline and project emissions
Calculation method	Based on sample-based monitoring, the parameter is calculated as: $n_{j,k,y} = q_{j,k,y} \times \frac{n_{op,j,k,y}}{n_{tot,j,k,y}}$ where n_{op} = number of operational ICS monitored n_{tot} = total number of ICS monitored
Comments	--

Data / Parameter	Hh_i $Hh_{j,k}$
Data unit	Equivalent standard male adults
Description	Average household size of the target population using baseline device type i Average household size of the target population using project device type j from batch k
Source of data	Baseline and Project KPT surveys (See “ICS-01 KPT Instance 1”)
Description of measurement methods and procedures to be applied	During KPT surveys, the household composition of families is recorded. The equivalent standard male adult per household is then identified, as per <i>Guidelines for Woodfuel Surveys for FAO</i> referenced in the methodology, and an average taken to find the average household size.
Frequency of monitoring/recording	New values are determined every time the project activity is undertaken in a new instance
Value applied	For the first project instance: $Hh_i = 3.38$ (includes control households) $Hh_{j,k} = 3.45$
Monitoring equipment	Surveys using tablets with the Tramontana ICS software application
QA/QC procedures to be applied	The value obtained has been compared to the average household size in the region as reported by the latest official government census. The project aims to exceed the required sample confidence threshold through using techniques such as oversampling.
Purpose of data	Estimation of average energy consumption per household, and for cross-checking of energy and fuel consumption values
Calculation method	For each family, the household composition of families is recorded, converted into the equivalent male adult units as per the Kitchen Performance Test Protocol v4.0, and finally averaged across all KPT households
Comments	This value is specific for Instance 1 and will be measured for each individual instance

Data / Parameter	n_{HH}
Data unit	Number
Description	Number of ICS distributed to each household
Source of data	Distribution database
Description of measurement methods and procedures to be applied	When an ICS is distributed by a field team member, the QR code is scanned and a corresponding family record is created and synced into the distribution database. Each family record is associated with a unique BPL card (see section 5.3.1). When there are multiple ICS records linked to a single BPL card, it signifies that multiple ICS have been distributed to a single family. Generally if a family has six or more members, they may request a second stove. This parameter is not calculated on a sample basis, but instead on the total project population. Thus it does not form part of sample-based monitoring in section 5.3.2.
Frequency of monitoring/recording	At the time of commissioning/distribution of ICS
Value applied	1 as standard. If a family requests more than one ICS, the Project Proponent will consider providing an additional ICS if, amongst other additional details obtained, the amount of firewood used is measured to be significantly higher than the average across the mandal.
Monitoring equipment	Surveys using tablets with the Tramontana ICS software application
QA/QC procedures to be applied	--
Purpose of data	Calculation of baseline and project emissions
Calculation method	--
Comments	--

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of woody biomass that is established to be non-renewable used by device in year y

Source of data	Option 2 as specified in the VM0050 v1.0 methodology
Description of measurement methods and procedures to be applied	Calculated by determining the share of non-renewable woody biomass in the total quantity of woody biomass consumed for each project instance state. The parameter value is obtained by using CDM Methodological Tool 30³⁰ with inputs specific to the local state of the project instance. An uncertainty discount of 26% is then applied to the calculated value. From survey results under the first instance it was recorded that users primarily sourced firewood from forests, which has led to widespread deforestation across the region (as shown for example in Figure 8). For the Instance 1 a value of 0.510 is calculated – please refer to the spreadsheet “ICS-01 fNRB Calculation Instance 1”
Frequency of monitoring/recording	Determined ex-ante for a project Instance and fixed for the crediting period
Value applied	0.510
Monitoring equipment	The value for average consumption of wood fuel per household has come from the KPT tests described in section 5.3. All measurements have been taken by extensively trained field teams who follow KPT Protocol v4.0 and use digital scales to directly record the weight of wood. Moisture is also accounted for through the use of digital moisture meters. Other inputs to the parameter calculation have been obtained from the IPCC Guidelines for National Greenhouse Gas Inventories²⁵, the 2019 India State of Forest Report, the 2019-21 Andhra Pradesh National Family Health Survey and 2011 Census of India Population Projection. Note that all references and calculations are described in detail in the spreadsheet “ICS-01 fNRB Calculation Instance 1”.
QA/QC procedures to be applied	To ensure the reliability of <i>NRB</i> and <i>RB</i> inputs, data has been crosschecked with other sources and publications available. These crosschecks include research from university studies as well as government publications. These are also presented in the spreadsheet “ICS-01 fNRB Calculation Instance 1”.
Purpose of data	Calculation of baseline and project emissions

³⁰ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf>

Calculation method	The value as per CDM Methodological Tool 30: Calculation of the fraction of non-renewable biomass³⁰ is calculated using: $f_{NRB} = \frac{NRB}{NRB + RB} \times (1 - 0.26)$ where NRB is the quantity of non-renewable biomass consumed in tonnes and RB is the quantity of renewable biomass in tonnes in the area. The factor 26% is applied under the latest methodology.
Comments	--

Data / Parameter	$BC_{b,i,y}$
Data unit	Tonnes
Description	Average quantity of fuel per baseline device type i during year y
Source of data	Option 1 as specified in the VM0050 v1.0 methodology
Description of measurement methods and procedures to be applied	The parameter is measured by conducting in-person KPTs in sample households for each project instance. Each test lasts 4 days and, in line with the methodology, the result is scaled using the average household size. For each project Instance, a minimum of 50 households will remain outside of the project as “control households”. If, throughout the project lifetime, it is observed that fuels, fuel sources or technologies used by these households undergo any changes, the baseline consumption value used for emission reductions calculations will be updated. The parameter is measured separately for each instance to gauge an accurate representation of fuel use in the specific region
Frequency of monitoring/recording	Every five years
Value applied	For the first project instance in Andhra Pradesh, an average value of 13.22kg per day was measured, equivalent to 4.83 tonnes annually (See “ICS-01 KPT Instance 1”)
Monitoring equipment	Field teams directly record the weight of the wood using digital scales. Moisture is also measured using a digital moisture meter.

	The digital scales used by the field team are the ATOM A 302 Electronic Digital Hanging Stainless Steel Hook Luggage Portable Scale with resolution 0.01kg and accuracy +/-0.01kg The moisture meters used are the Signette-MT10-R-1 with resolution 0.1% and accuracy +/-0.5%. All measurements are recorded on to the Tramontana ICS application and stored in the cloud database
QA/QC procedures to be applied	The digital moisture meters and weighing scales have been calibrated by the manufacturer to help ensure accuracy of measurements taken. Field staff conducting the tests are given KPT specific training ensuring the accuracy and reliability of the results collected The test results from each batch must meet the 90/10 confidence/precision as set by the methodology, else more sample tests are undertaken until the requirement is met. This is discussed further in section 5.3.
Purpose of data	Calculation of baseline emissions
Calculation method	--
Comments	The parameter is <u>dynamic</u> and is chosen to be measured separately for each instance in order to obtain an accurate representation of fuel use in the specific region.

Data / Parameter	$\eta_{new,avg,y}$
Data unit	Fraction
Description	Weighted average efficiency of project devices in year y
Source of data	Option 1 as specified in the VM0050 v1.0 methodology
Description of measurement methods and procedures to be applied	The parameter is measured by conducting Water Boiling Tests (WBT) involving averaging the cold-start and hot-start high-power phases of the ICS. In line with the methodology, for an ICS of type i and vintage y a sample from the first ICS batch of 3 ICS are selected and for each one 3 WBTs are conducted. The degradation of efficiency measured from monitoring the first ICS batch may be applied across all batches. The WBT is conducted by an independent technical laboratory and certified test results will be provided as part of monitoring
Frequency of monitoring/recording	Annually

Value applied	For y=0, the thermal efficiency of the CRS-23 model project stove measured from tests conducted on 3 randomly selected operational ICS resulted in an average value of 36.01% See Appendix 1
Monitoring equipment	Laboratory tests
QA/QC procedures to be applied	Experienced technical laboratories following ISO standards are selected for the purposes of conducting all WBTs. The test results from the samples must meet the standard deviations limit and 90/10 confidence/precision as set by the methodology, else more sample tests will be undertaken until the requirement is met. This is discussed further in section 5.3.
Purpose of data	Calculation of baseline and project emissions
Calculation method	For an ICS of type i and vintage y, the parameter is determined by averaging the results from independent WBTs conducted on a sample of a minimum of 3 operational ICS
Comments	--

Data / Parameter	$BC_{p,j,k,y}$
Data unit	Tonnes
Description	Average quantity of fuel used by project device type j from batch k during year y
Source of data	Option 1 as specified in the VM0050 v1.0 methodology
Description of measurement methods and procedures to be applied	The parameter is measured by conducting in-person KPTs in sample households participating in the project activity. Each test lasts 4 days and, in line with the methodology, the result is scaled using the average household size. The tests are also conducted in paired samples with the baseline KPTs, so that a direct comparison of pre-project and post-project consumption can be made for each household.
Frequency of monitoring/recording	At least biennially
Value applied	For the first project instance in Andhra Pradesh conducted in January 2025, an average value of 4.80kg per day was obtained for each Tramontana ICS, which is equivalent to 1.75 tonnes annually

Monitoring equipment	Field teams directly record the weight of firewood using digital scales. Moisture is recorded using a digital moisture meter. The digital scales used by the field team are the ATOM A 302 Electronic Digital Hanging Stainless Steel Hook Luggage Portable Scale with resolution 0.01kg and accuracy +/-0.01kg The moisture meters used are the Signette-MT10-R-1 with resolution 0.1% and accuracy +/-0.5%. All measurements are recorded using the Tramontana ICS application and stored in the cloud database
QA/QC procedures to be applied	The digital moisture meters and weighing scales have been calibrated by the manufacturer to help ensure accuracy of measurements taken. Similar to baseline KPTs, field staff conducting the tests are given KPT specific training ensuring the accuracy and reliability of the results collected. The test results from each batch is to meet the 90/10 confidence/precision as set by the methodology, else more sample tests will be undertaken until the requirement is met. This is discussed further in section 5.3.
Purpose of data	Calculation of baseline and project emissions
Calculation method	--
Comments	--

5.3 Monitoring Plan

5.3.1 Data Recording

The Project Proponent will obtain the necessary information to identify households using an ICS during the course of the project. To facilitate this process, the Project Proponent has specifically developed the Tramontana ICS software application which is its own proprietary application on the Android operating system (OS) for conducting surveys. The application runs on portable tablets which are used by trained local field staff specifically employed for the project. Each ICS (see details in Tables 2-5) is assigned a unique serial code which is displayed on the device in writing and in the form of a QR code. For a new end user record as part of the registration process, the QR code is scanned using the Tramontana ICS software application and the following information is recorded. The information includes (as appropriate and as available):

- Name, phone number and photo of beneficiary/head of household with their ICS
- Address of ICS household
- Statutory Community Classification (if any)

- Pre-project cookstove type
- Photos of ID card and other certification
- ICS model
- ICS serial number
- GPS coordinates of ICS household
- Date and time of ICS installation
- Signed ICS licence agreement (including credit ownership)
- On-field distribution team member information including name, title and signature

Once a new user record is created in the Tramontana ICS software application it is automatically synced to a Google cloud database once the tablet is brought online in the local office. Figure 16 is a screenshot of a section of the app used for registration of new ICS beneficiaries. All information collected/recorded including for baseline, registration and monitoring purposes is stored in the online database and subsequently automatically mapped into an Excel sheet. This sheet serves as the basis for monitoring and auditing throughout the project life.

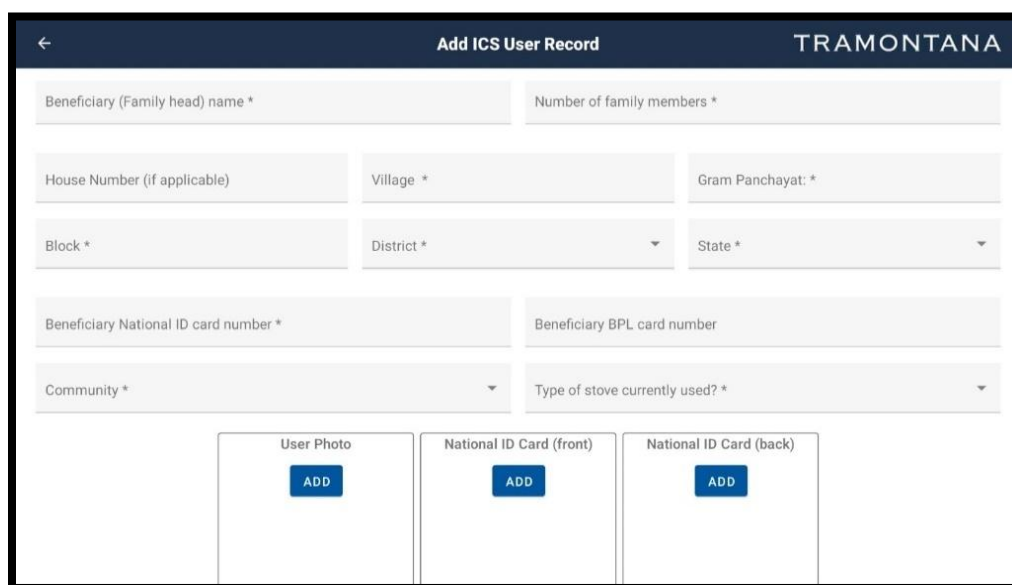


Figure 16. Screenshot from the new ICS user record section of the Tramontana ICS software application

5.3.2 Sampling

A sampling plan is required for the purposes of monitoring and verification of project emissions reductions. As per CDM guidelines³¹ a sampling plan specifically contains (i) a sample design/sizing and field data and (ii) an implementation plan and data quality control.

³¹ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20151023152925068/Meth_GC48_%28ver04.0%29.pdf

5.3.2.1 Sample design/sizing and field data

The target population includes all households under the grouped project who are having their traditional stoves replaced by a new Tramontana ICS. Owing to the large number of ICS planned to be distributed under the project, representative sampling of the population is applied, in line with the CDM sampling guidelines.

The objective of sampling is to obtain an unbiased statistically significant estimate of the lifecycle variables to be monitored from the population, presented in Table 11, for the purposes of emissions reduction calculations over the course of the project period. The parameter type column specified in the table corresponds to either proportion or mean value as per the methodology guidelines.

Table 11. Parameters which are sampled and monitored through the project lifecycle

Parameter	Type
$n_{j,k,y}$ Proportion of commissioned project devices of type j in batch k operating during year y	Proportion
$BC_{b,i,y}$ Average quantity of fuel used per baseline device type i during year y	Mean Value
$BC_{p,j,k,y}$ Average quantity of fuel used by project device type j from batch k during year y	Mean Value
$Hh_i / Hh_{j,k}$ Average household size of the target population using baseline device i / project device j from batch k	Mean Value
$\eta_{new,avg,y}$ Weighted average efficiency of project devices in year y	Mean Value

The parameter $\eta_{new,avg,y}$ in Table 11 is monitored using the approach outlined in the Water Boiling Test Protocol v4.2.3³² in compliance with CDM Standard for Sampling and Surveys³³. More

³² <https://cleancooking.org/binary-data/DOCUMENT/file/000/000/399-1.pdf>

³³ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf

specifically, an average is calculated from a minimum of 9 samples (3 ICS with 3 WBTs conducted for each). A simple random sampling approach described below is applied to identify the specific ICS to be tested for the respective monitoring period. Note that samples are taken entirely from the first batch of an ICS type, and the corresponding efficiency values obtained for each ICS vintage (i.e. forming an efficiency curve) are applied for all subsequent project ICS distributed of that type. Separately, as the WBT sample size to be used is less than 30, the Student's t-distribution is applied to ensure a 90/10 confidence/precision level is met. If the requirement is not met, more samples will be taken until the threshold is achieved. Note for $y=0$, the manufacturer-certified WBT value has been used (option 2 from the VM0050 v1.0 methodology) and therefore statistical analysis has not been conducted on the parameter. See Appendix 2 for test results.

For the remaining four parameters in Table 11, sampling is conducted separately and independently for each project activity instance under the grouped project, to improve data reliability and accuracy. Note the sampling frame is taken as the set of all information recorded in the cloud database for a given project instance.

There is no necessity to account for different ICS vintages when sampling, given distribution for a single project instance is unlikely to occur over many years. It should also be noted that geographical variations have minimal influence on the monitored parameters within an instance. Therefore owing to the homogenous nature of an instance population, a Simple Random Sampling technique is used for sampling.

Note that if for any reason a new ICS model or variation on an existing ICS model is included into the project in the future, it will be grouped into a separate population unit and the sampling approach described in this section will be applied to it.

To undertake a Simple Random Sampling approach for the parameter $n_{j,k,y}$, a random number generator must be utilised. Given each ICS has a unique serial code identifier, we can first pair each ICS with a sample number. For instance, the samples can begin from 1 and increase up to the total population size of the instance corresponding to all successfully distributed ICS. The random number generator can then be applied to randomly select an ICS from the population. The task is repeated until the targeted sample size is achieved.

The first four monitored parameters in Table 11 are determined through field surveys conducted by the team. More specifically, for each of the parameters:

$BC_{b,i,y}$	KPT is conducted in-situ over a four day period with in-person visits to the households using a pre-project baseline stove
$BC_{p,j,k,y}$	KPT is conducted in-situ over a four day period with in-person visits to the households using a project device
$Hh_i / Hh_{j,k}$	Determined during KPTs where the gender and age of all residents is recorded, and applied under the Kitchen Performance Test Protocol v4.0 to calculate the standard male adult equivalent per household

$n_{j,k,y}$ A site visit is carried out to inspect the household and identify whether the ICS is in operation including taking photos of the stove, its components and the cooking area. A detailed survey is also conducted with the end user to identify current usage patterns and operation/performance over the past year

In addition to covering these monitored parameters, monitoring surveys provide a wider and more comprehensive review of recent ICS usage, performance and perceptions by end users. Photographs are also taken of the beneficiary, their Tramontana ICS and cooking area all of which are provided to Verra and the VVB for review.

For each of the parameters in Table 11, further details are provided below on how they are monitored and sampled in line with the CDM Standard for Sampling and Surveys.

The parameters $BC_{b,i,y}$ and $BC_{p,j,k,y}$ are monitored using the approach described in the Kitchen Performance Test Protocol v4.0³⁴ and the methodology. Specifically a paired-sample study is conducted, first with households in the baseline scenario and then in the post-distribution project scenario. The use of the same households allows direct comparison of fuel use. We also keep separate “control households” to reflect any changes in baseline characteristics over time. This corresponds to households that remain outside of the project (i.e. not receiving a Tramontana ICS), with a minimum 50 such households maintained in each instance area.

To select the households for surveying, a simple random sampling approach is used. Specifically, each household from the survey is assigned a unique identifier. Using Excel's RANDARRAY function, a sample of random unique identifiers is then generated, representing a sample of the population. This approach guarantees that each household has an equal chance of being selected, resulting in a representative and unbiased sample.

Each of the KPTs last four days and is typically conducted Monday-Thursday to avoid weekends. Festivals and other holidays where there may be above-average fuel consumption are also avoided. The test results from the households are required to meet a 90/10 confidence/precision as set by the methodology, else more sample tests will be undertaken until the requirement is met. For the first project area, Instance 1, the KPT measured $BC_{b,i,y} = 4.83$ tonnes/year. From the 151 samples (n) taken, the mean and standard deviation (SD) of the data is calculated as 3.95 kg/pp/day and 0.99 kg/pp/day respectively (see “ICS-01 KPT Instance 1”). From these parameters, the confidence interval (CI) and precision are given as follows:

$$CI = mean \pm 1.645 \times \frac{SD}{\sqrt{n}}$$

Note that 1.645 represents the 90% confidence required by the VM0050 v1.0 methodology.

The upper and lower bound of CI are calculated and the difference is outputted to give:

$$CI = 3.95 \pm 1.645 \times \frac{0.99}{\sqrt{151}}$$

³⁴ <https://cleancooking.org/binary-data/DOCUMENT/file/000/000/604-1.pdf>

$$\Delta CI = 0.265$$

$$Precision = 0.5 \times \frac{\Delta CI}{mean}$$

$$= 0.5 \times \frac{0.265}{3.95}$$

$$= 3.35\%$$

As the precision is less than 10%, the mandatory 90/10 rule has been met.

The parameter $Hh_i / Hh_{j,k}$ is also monitored during the KPT surveys. Specifically, household composition is recorded and the equivalent male adult per household calculated as per the Kitchen Performance Test Protocol v4.0. A mean is then taken across households to determine the average household size. To ensure the parameter Hh_i (and baseline survey outputs) also meet the 90/10 confidence/precision requirements from the methodology and CDM guidelines, the mean and standard deviation are calculated from the 151 samples (n) taken, as 3.38 and 1.25 respectively. Following the same approach used above, ΔCI and precision are calculated as follows:

$$CI = 3.38 \pm 1.645 \times \frac{1.25}{\sqrt{151}}$$

$$\Delta CI = 0.334$$

$$Precision = 0.5 \times \frac{0.334}{3.38}$$

$$= 4.95\%$$

As the precision is less than 10%, the mandatory 90/10 rule has been met and therefore the parameter Hh_i (and baseline survey outputs) for Instance 1 achieves the requirements of the methodology and CDM guidelines.

For the final parameter in the table above $n_{j,k,y}$, in line with the *Correction and Clarifications to VM0050*²⁷ document published in February 2025, it is capped at 0.9 as long as the following actions are undertaken:

1. *Selection of technologies and fuels that fully meet the cooking needs of the target population, demonstrated by citing robust research or conducting an investigation of cooking practices and attitudes*
2. *Implementation of support activities to assist the target population in effectively operating and maintaining their cookstove. These may include providing and maintaining their cookstoves. These may include providing materials (print, in-person, or video) on how to operate the cookstove to prepare common local foods, how to troubleshoot common operational issues, and how to make minor repairs (including obtaining necessary replacement parts). All such communications and materials must be provided in local language(s) commonly used in the project area*

3. *Provision of a commonly used, toll-free communications channel through which the target population can contact the project proponent to access support (e.g. maintenance and repair services)*

The Tramontana ICS has been specially designed to meet all cooking requirements of the target population. This includes for food preparation, tea/coffee preparation and water boiling purposes for bathing and drinking. Extensive baseline studies conducted in the project area found that households traditionally use a single mud-clay stove, fuelled by wood to meet their cooking needs. The Tramontana ICS serves as a direct replacement for these traditional stoves and uses the same fuel type, which promotes consistent adoption and minimises use of alternative stoves, thus satisfying criteria 1) above. Monitoring surveys carried out throughout the duration of the project will verify usage patterns and record any deviations from the intended stove use.

To ensure the sustained use of the ICS and satisfy criteria 2) above, the Project Proponent has implemented multiple initiatives. At the point of ICS distribution, households receive a brochure, written in the local language, which includes instructions on how to use and maintain their ICS. In case of any stove issues, the Project Proponent has also employed full-time dedicated maintenance teams who carry out repairs and replacements at no cost to end users. Replacement parts are stored locally in field offices enabling swift repair work thanks to the readily available components. This approach ensures that households can continue using their ICS without disruption throughout the project's duration. In addition to the maintenance teams, local staff conduct routine household visits to offer guidance on best practices and provide tips. All such staff are from the local communities and therefore speak both the local language and dialect, fostering strong relationships and trust.

During distribution, access to a local staff member and their phone number is provided to every end user in case of any issues faced with their ICS. This ensures that the Project Proponent can immediately resolve any issues as they arise. Therefore criteria 3) above is met.

From the justifications provided above, it is clear that all three criteria are met and therefore we adopt the capped usage rate for $n_{j,k,y}$ of 0.9.

5.3.2.2 Sample size

In line with the methodology, a minimum 90/10 accuracy is to be achieved for all sampled parameters.

As $n_{j,k,y}$ is a proportion parameter, the minimum sample size is formulated using:

$$n \geq \frac{1.645^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.645^2 \times p(1-p)} \quad \text{Equation 15}$$

Where:

n = Sample size for households

N = Total number of households

p	=	Expected proportion of parameter of interest
1.645	=	Represents the 90% confidence required
0.1	=	Represents the 10% relative precision

To perform sample size calculations, the parameters in Table 11 are first estimated. Following the methodology, this is achieved through one of the following:

1. *Preliminary data analysis: conduct an exploratory study using previously collected data, where available. This can give a first idea of the data distribution and parameter variability.*
2. *Literature research: previous research related to the topic of study can be examined.*
3. *Pilot or feasibility studies: conduct pilot or feasibility studies to gather preliminary data. This can help estimate parameters of interest and adjust sample size calculations accordingly.*
4. *Expert consultation: obtain opinions and knowledge from experts in the relevant field.*
5. *Own experience: using estimates based on researcher's own experiences.*

For registration purposes, option 5 is used. Note we select a value of $p = 0.6$ for $n_{j,k,y}$, which is conservative as the Project Proponent is continuously repairing/replacing all ICS and all traditional cookstoves are aimed to be demolished/removed. For subsequent monitoring periods, option 1 above could alternatively be followed to estimate p through a value determined from the previous monitoring period.

Taking the case of the first project activity instance where the ICS is distributed in the ASR district of Andhra Pradesh, it is estimated that the total number of beneficiary households (population N) will be around 50,000.

Applying Equation 15 denoting Simple Random Sampling, we calculate the sample size for a 90/10 precision as:

$$n \geq \frac{1.645^2 \times 50,000 \times 0.6(1 - 0.6)}{(50,000 - 1) \times 0.1^2 \times 0.6^2 + 1.645^2 \times 0.6(1 - 0.6)}$$

$$n \geq 179.76$$

$$n = 180$$

Therefore the minimum sample size for monitoring as part of the first monitoring period is set to 180. In line with the Project Developer's high standards, significant oversampling is applied as detailed in the corresponding monitoring report.

For the remaining three parameters in Table 11, $BC_{b,i,y}$ and $BC_{p,j,k,y}$ and $Hh_i / Hh_{j,k}$, a different approach is used as these are mean parameters. In line with the methodology, to determine the sample size for a mean parameter the following equation is used:

$$n \geq \frac{1.645^2 N \times V}{(N - 1) \times 0.1^2 + 1.645^2 \times V}$$

Equation 16

$$V = \left(\frac{SD}{mean} \right)^2$$

Where:

n	=	Sample size for households
N	=	Total number of households
SD	=	Expected standard deviation
$mean$	=	Expected mean, depends on similar studies or location
1.645	=	Represents the 90% confidence required
0.1	=	Represents the 10% relative precision

From this data, we can calculate a sample size for the KPTs which determine $BC_{b,i,y}$, $BC_{p,j,k,y}$ and $Hh_i / Hh_{j,k}$. First the baseline fuel consumption per person per day is evaluated, for which a standard deviation and mean can be estimated. Using an expected population (N) in Instance 1 of 50,000, we calculate a mean and standard deviation of 5.12 kg/pp/day and 1.85 kg/pp/day respectively. Note these values were estimated using preliminary data on baseline practices across 1793 households surveyed in the Instance 1 region.

Equation 16 can then be applied to calculate the minimum sample size:

$$V = \left(\frac{1.85}{5.12} \right)^2$$

$$V = 0.13 \text{ (2 d.p.)}$$

$$n \geq \frac{1.645^2 \times 50,000 \times 0.131}{(50,000 - 1) \times 0.1^2 + 1.645^2 \times 0.131}$$

$$n \geq 35.33 \text{ (2 d.p.)}$$

$$n = 36$$

To increase the reliability of our results, in Instance 1 we have conducted 4.2 times the number of KPT surveys as required by the minimum sample size, and expect to do similar across all Instances.

In line with section 6.2 of VM0050, baseline surveys must also be conducted following the CDM Standard for Sampling Guidelines, and therefore a minimum sample size of 36 is required to meet the 90/10 confidence/precision level. For the first project Instance, 1,994 baseline

surveys have been conducted. This is 56 times the minimum required, demonstrating a significant level of oversampling.

5.3.2.3 Implementation plan & data quality control

The Project Proponent will oversee and manage the monitoring process and sampling plan to ensure accurate measurement and reporting of emissions reductions from the project activity throughout the project life. The methods set out in Table 12 are applied for the purposes of monitoring the parameters from Table 11, and this is undertaken with representative sampling as outlined earlier in this section to identify the households to be included in the respective monitoring cycle.

Table 12. Frequency and methods with which project parameters are monitored

Parameter	Monitoring frequency	Frequency as required by methodology	Monitoring methods to be applied	Seasonal fluctuation
$n_{j,k,y}$	At least annually	Annual	Site visits to undertake visual examination and conducting detailed surveys/interviews	None
Hh_i $Hh_{j,k}$	For each project instance and monitored at least the frequency required by the methodology	Every 5 years for Hh_i Biennial for $Hh_{j,k}$	Family composition is recorded during baseline surveys	None
$BC_{b,i,y}$	Determined for each project instance and monitored at least every 5 years	Every 5 years	KPTs conducted by trained field staff prior to ICS distribution in households	None
$BC_{p,j,k,y}$	Determined for each project instance and monitored at least biennially	Biennial	KPTs conducted by trained field staff in project households	None

$\eta_{new,y,i}$	Annual	Annual	WBTs conducted by independent technical laboratories	None
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Substantial oversampling is applied for the first four parameters in Table 12 in order to enhance the reliability of the data recorded, as well as to adjust for any errors or cases of answer bias that may be observed. Note for $n_{j,k,y}$ outliers do not exist as it corresponds to proportions as opposed to mean values under the methodology.

In addition to oversampling, an independent set of KPTs were conducted in-situ by a specialist external company Koshish Sustainable Solutions, with vast experience of biomass cookstoves testing both lab and in-situ. Dr Amit Ranjan Verma (PhD – IIT Delhi) from the firm and his team: Dr Ratnesh Tiwari (PhD – IIT Delhi) and Vishnu Chatriya (BA – Rajendra College Chapra), conducted KPTs in Instance 1 between the dates of 15th December 2024 and 21st December 2024 with the baseline stove, and then later returned between 20th January 2025 and 25th January 2025 to perform KPT tests on the same households after they received a Tramontana ICS. The study found an average of 14.5kg of wood per household per day was being used in the baseline scenario, which is in line with the 13.22kg measured from our KPTs.

To successfully implement the project, an extensive training programme is conducted prior to distribution in each instance, for all local field staff and coordinators employed for the purposes of data collection. This includes classroom lectures (as shown in Figure 17) outlining survey structure, purpose and goals, demonstrations of the Tramontana ICS software application (see Figure 16), as well as on-field practice sessions to gain experience in using the tablets containing the application. Additional training sessions are also provided for field staff conducting baseline, monitoring and KPT surveys where specialist techniques are required, such as the use of digital moisture meters or to follow the KPT protocol. Regular reviews are carried out for all staff and new employees are supervised by management to ensure full understanding of their role before beginning any project work.

A key additional aspect of the training programme is to convey engagement techniques with the community, approaches to minimising errors, and methods to deal with issues faced such as non-response cases, conflicts of interest and interviewer bias. Members of the local community are typically employed as field staff in order to maintain closer community engagement and maximise response rate, whilst it also provides support to the local people in the form of employment and training. All field staff within a mandal are accompanied by an experienced mandal coordinator who helps oversee activities, ensuring the team works in an efficient manner and to help tackle any problems faced.



Figure 17. An internal classroom training session held for field team members in advance of ICS distribution

All surveys used for the project have been designed and tested by the Project Proponent in consultation with the various stakeholders. The surveys have been programmed into the Tramontana ICS software application which is used by the field teams. The team are able to view the application on tablets, which have been purchased specifically for the purposes of the project activity. The application has been developed to work in an offline mode given the remoteness of project households/beneficiaries. Once a field team member has successfully completed a survey (after meeting all requirements and passing the built-in checks in the application), they can submit the entry on the tablet to create a record for the respective beneficiary. The built-in checks in the application such as mandatory fields/data types minimise any incorrect or incomplete surveys from being submitted. The submitted record is stored locally and is then automatically synced to a Google cloud database once the tablet is brought online under an internet connection in the local office. This automation of syncing maintains the data integrity and minimises any risk of data loss, theft, tampering/fraud or negligence/errors in translation. Once the data is in the cloud database, it is automatically processed and translated into an Excel sheet format which can be easily analysed by the VVB, Verra and any other project parties.

Once distribution/monitoring data has been recorded, submitted and translated into the Excel sheet, it is checked by the dedicated office team based in Hyderabad for the purposes of quality control against a pre-defined set of criteria developed by the Project Proponent. The team's tasks include checking all: written responses, photos (including their content, clarity and positioning), signed licence agreements and location coordinates. Any deviation from the set of criteria is marked, rejected, and immediately sent for resurvey upon coordinating with the local field offices. Incorrect surveys are later removed from the database once the resurveys have been successfully completed. The correct entries are then sent to the London-based office team who undertake further analysis, including data processing, compilation and statistical analysis for certification purposes. Both London and Hyderabad teams engage in daily communications to ensure any issues or bottlenecks are addressed promptly and to provide support where needed, reflecting an effective and coordinated working relationship.

Prior to any documentation or data being shared with a VVB or Verra, it goes through a set of approvals and requires sign off by Director level personnel at the company.

Version control mechanisms are employed by both office teams. Using Microsoft SharePoint, the teams utilise version control through generating new document versions whenever edits are made and saved in a document library. This system enables the Project Proponent to view detailed version histories, compare edits, and revert to earlier versions when necessary. In addition, new versions are intentionally created before making any major edits to ensure important changes are tracked with clarity. Older versions are retained and not deleted, providing a full archive for reference, accountability, or rollback if required. Versioning also ensures that any errors or updates are logged to maintain transparency within all documents provided.

Note that at the initial stage after ICS distribution some additional checks are also carried out. This includes a separate dedicated team calling and checking all phone numbers in the database that were provided by households during distribution. This is to ensure beneficiaries are reachable if needed at any time by the Project Proponent.

Any ICS that is found not to be in use post-distribution or from a monitoring survey will not be counted towards the total number of ICS in operation for the preceding period or thereafter. Further, any use of alternative stoves, for instance which may have been built or purchased since the distribution of the project ICS, will be accounted for using the method described in the table in section 5.2 for the monitored parameter $n_{j,k,y}$.

The final data available in the online database will include (i) initial information obtained during ICS distribution (outlined in section 1.12.1) and (ii) monitoring data recorded (including KPT results) from monitoring surveys (and laboratory tests) as outlined in section 5.2. The data will be extensive and directly attributable to a particular ICS/household, forming clear evidence for any emissions reductions that are claimed.

For the purposes of monitoring, the Project Proponent will compile a regular monitoring report to be submitted to the VVB as part of project audits throughout the project life. This will detail the monitoring activity and monitoring data recorded in the online project database and from laboratory tests, as well as the associated calculations specifying the emissions reductions from the project activity for the respective period.

APPENDIX 1

The schedule followed including the event details for stakeholder meetings conducted for the first project instance in Andhra Pradesh is outlined.

Mandal	Village	Meeting Venue	Attendees	Date	Special Guests
Paderu	Modapalli	AMC Yard	105	14-Nov-2022	Kottagulli Usha Rani (President (Sarpanch) of GP Paderu) Sonari Ratnakumari (Mandal Parishad President of Paderu) Sonari Venkata Ramana Murthy (ex-Mandal Parishad President of Paderu)
	Onuru	Anganwadi Centre	52	15-Nov-2022	Thokala Eswara Rao (MPTC of Vantalamamidi) Pangi Rambabu (President of Vantalamamidi) Palliki Lakku (CFFI National Member)
	Paderu	Sri Modhakond ammathalli Temple	310	28-Dec-2022	Sri V. Abhishek (Sub Collector of A.S.R. district) Sri Shathakabulli Babu (Tricor Chairman of Paderu) Kotagulli Usharani (President of Paderu) Rathnakumari (MPP of Paderu) Sri Ramana Murthy (Former MPP of Paderu)
G. Madugula	G. Madugula	Rachabanda	108	16-Nov-2022	Mathyarasa Gayathri Devi (Market Yard Chairman of G. Madugula and Paderu mandals) Kimudu Rambabu (President of G. Madugula)

					Mathyasara Vijaya Kumari (MPTC Member of G. Madugula) Pangi Chitti Babu (MPTC of G. Madugula)
	Gaduthuru	Anganwadi	137	17-Nov-2022	Kondapalli Simhachalam (President of Gaduthuru) Madhusudhan Babu (Secretary of Gaduthuru GP) K. Chandrapathi (President of PESA (Panchayaths Extension to Scheduled Area) of Gaduthuru)
Hukumpeta	Dabalavasa	Jio Tower	79	20-Nov-2022	Korra Raja Rao (Former MPTC of Hukumpeta) Kincheyi Krishna (President of Dabalavasa)
Pedabayalu	Galaganda	Panchayath Office	82	19-Nov-2022	Pujari Lohidasu (President of Galaganda) Thalla Bullamma (Former President of Galaganda) Marri Devayya (Former President of Pedakodapalli)
	Pedabayalu	Girijana Bhavan	63	21-Nov-2022	Korra Raju Babu (Vice MPP of Pedabayalu) Vanthala Ananda Rao (MPTC of Pedabayalu) Gullili Krishna Rao (President of Pedabayalu)
Chinthapalli	Gondhipakalu	Grama Sachivalayam	102	22-Nov-2022	Thambeli Mohan Rao (MPTC of Chinthapalli) Sagila Varalaxmi (President of Gondhipakalu)
	Yerrabandha	GCC Godowns	103	24-Nov-2022	Segge Sathibabu (MPTC of Chinthapalli)

					Lotha Pandayya (President of Pedhuru) Kuda Chinnabbai (Former President of Pedhuru) Kuda Raju Babu (VRP NREGS of Pedhuru)
G.K. Veedhi	Devarapalli	Grama Sachivalayam	118	23-Nov-2022	Pothuraju Krishna Murthy (MPTC of Devarapalli) Siribala Bujji Babu (President of Devarapalli) Matchala Mathya Raju (Former ZPTC of Devarapalli)
	Kothapalem	MPP School	102	25-Nov-2022	Motanam Sathyanarayana (MPTC of G.K. Veedhi) Kankipati Veerendra Prasad (President of Kothapalem village) Kankipati Ramarao (Former MPTC of G.K Veedhi)
Araku Valley	Baski	Church Ground	107	26-Nov-2022	Ranjapalli Usharani (MPP of Araku Valley) Padi Ramesh (President of Baski) Killo Ramanna (Vice MPP of Baski)
	Nandi Valasa	MPP School Building	84	28-Nov-2022	Swabi Ramachandra (MPTC of Madhala) J. Kiran Kumar (Panchayath Secretary of Madhala village)
Ananthagiri	Venaganda	MPP School Building	53	27-Nov-2022	Sembi Sanyasi Rao (President of Vaalasi) Sodipalli Seema (President of Venaganda) Siveri Kondala Rao (Former President of Venaganda)

Munchingiputtu	Munchingiputtu	Community Hall	161	30-Nov-2022	Jallipalli Subhadra (Z.P chairperson of Munchingiputtu) P.Sathyanarayana (MPP of Munchingiputtu) K.Kiran Kumar (MPDO of Munchingiputtu) Seesa Subhash (President of Kinchayiputtu)
	Maddulabanda	MP Primary School	65	02-Dec-2022	K. Muktamma (President of Kamada GP) K. Radhakrishna (G.P.S School Teacher in Kamada GP)
Dumbriguda	Dumbriguda	IKP Office	197	01-Dec-2022	V. Nagesh (MPDO of Dumbriguda) M. Geetha (MPTC of Dumbriguda)
	Parasilla	Rachhabanda	75	03-Dec-2022	Shobha Raghupathi (MPTC of Parasilla) Veeraboyina Subbarao (President of Parasilla)
Koyyuru	Yerri Naidu Pakalu	Near Community Hall	65	03-Dec-2022	Badugu Ramesh Babu (MPP of Koyyuru) Jampa Rajakumari (President of Koyyuru)
	Palakajeedi	Government School Building	63	05-Dec-2022	B. Ramesh Babu (MPP of Koyyuru) Dadala Ramesh (President of Palakajeedi) Thammarbha Madhava Rao (MPTC of Palakajeedi)

APPENDIX 2

Laboratory water boiling test results on the Tramontana ICS CRS-23 (detailed in Table 2 and Table 3) conducted by the Department of Mechanical Engineering at Jorhat Engineering College.

Thermodynamic Laboratory

Department of Mechanical Engineering

Jorhat Engineering College

Jorhat-7, Assam

www.jecassam.ac.in/



1. Water Boiling test results of FUELNZEL (CRS-23) stove

Fuelnzal (CRS-23), a natural draft biomass cook stove was tested in the Thermodynamic Laboratory of the department of Mechanical Engineering, Jorhat Engineering College, Jorhat. The stove was tested according to the water boiling test protocol WBT 4.2.3. The details of the stove and the test results are given below.

1.1 Details of FUELNZEL (CRS-23) Stove

Specification

Fuel Type	Wood
Feeding Type	Front Feeding
Combustion chamber	Ceramic
Cooking suitable for	5 to 10 persons
Net wet	11 kg
Draft	Natural draft

1.2 Test results

The average thermal efficiency of the stove was found to be 36%. The summary of the test results is given in the table below.

1. HIGH POWER TEST (COLD START)	units	Test 1	Test 2	Test 3	Average
Time to boil	min	39	37	37	37.7
Temp-corrected time to boil	min	38	36	36	37
Burning rate	g/min	9	10	10	10
Thermal efficiency	%	35%	35%	35%	35.1%
Specific fuel consumption	g/liter	77	77	75	76
Temp-corrected specific consumption	g/liter	74	75	74	74

Temp-corrected specific energy cons.	kJ/liter	1,318	1,330	1,315	1,321
Firepower	watts	2,752	2,899	2,844	2,831
2. HIGH POWER TEST (HOT START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot	min	34	37	36	35.7
Temp-corrected time to boil	min	33	36	36	35
Burning rate	g/min	11	9	9	10
Thermal efficiency	%	37.3%	36.9%	37%	37%
Specific fuel consumption	g/liter	77	73	71	73
Temp-corrected specific consumption	g/liter	75	71	71	72
Temp-corrected specific energy cons.	kJ/liter	,332	1,260	1,254	1282
Firepower	watts	3,118	2,743	2,770	2,877
3. LOW POWER (SIMMER)	units	Test 1	Test 2	Test 3	Average
BENCHMARK VALUES (for 5L)		Test 1	Test 2	Test 3	Average
Fuel Use Benchmark Value	g	373	364	361	366
Energy Use Benchmark Value	kJ	6,625	6,475	6,421	6,507
IWA PERFORMANCE METRICS	units	Test 1	Test 2	Test 3	Average
High Power Thermal Efficiency	%	36.4%	35.8%	36.0%	36.01%


17/01/2020

Tested by

D.K. Rabha
Assistant Professor
Department of Mechanical Engineering
Jorhat Engineering College
Jorhat, Assam

APPENDIX 3

Laboratory water boiling test results on the Tramontana ICS STAR-25 (detailed in Table 4 and Table 5) conducted by the School of Energy Science and Engineering at the Indian Institute of Technology.

*Final Report
on*

Testing of Thermal Efficiency of Improved Biomass Cookstoves *(A Consultancy Project)*

Sponsored by

Sustainable Energy Tech & Innovations Pvt Ltd
Address: D-148B, Ground Floor, Sector-8
Katihar, New Delhi-110085, India
Website: www.sustainableenergytech.com and info@sustainableenergytech.com

Project Investigators

Dr. Lepakshi Barbora	Senior Technical Officer, School of Energy Science and Engineering, IITG
Mr. Debarshi Baruah	Technical Officer (Gr-I), School of Energy Science and Engineering, IITG



School of Energy Science and Engineering
Indian Institute of Technology Guwahati,
November, 2024

Project Details:

Facilitating Organization: School of Energy Science & Engineering (Formerly, Centre for Energy), Indian Institute of Technology Guwahati, Assam-781039

Consultancy Project No:

Name of Consultants: Dr. Lepakshi Barbora, Senior Technical Officer, School of Energy Science and Engineering, IITG
Mr. Debarshi Baruah, Technical Officer (Gr-I), School of Energy Science and Engineering, IITG

Type of Equipment: Natural Draft Biomass Cookstoves

Model: Fuelnzl STAR-25

Design Reg. No.: 2023/24-001-0001

Name of Manufacturer: Fuelnzl, a subsidiary of Fuelnzl Pvt. Ltd.
Address: 10, 11th Floor, Fuelnzl House, Sector 10,
Indira Park, New Delhi-110028, India

Testing Period: November 2024

Name of Testers: Mr. Arup Dutta, Ms. Aparupa Thakuria & Mr. Moon Haloi

Objective of the Project: To evaluate the thermal efficiency of the cookstove following the Water Boiling Test (WBT 4.2.3) test protocol.

General specifications of the Stove (Fuelnzl STAR-25)

Fuel Type:	Wood
Draft:	Natural
Feeding Type:	Front feeding
Cooking suitable for:	5 to 10 persons (domestic)
Net Weight (Approx.):	8.65 kg
Dimensions:	275 (OD) × 386 (W) × 276 (H) mm



Fig. 1: Picture of Fuelnzl Star-25 stove Manufactured by Tira Montana Tech & Innovation Pvt Ltd

Equipment Testing

The mentioned Fuelnzl Star-25 stove was tested following Water Boiling Test (WBT 4.2.3) test protocol and the test results are as follows:

1. HIGH POWER TEST (COLD START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	26	20	30	25.33
Temp-corrected time to boil Pot # 1	min	27	21	31	26.33
Burning rate	g/min	9	12	9	10.00
Thermal efficiency	%	56%	52%	48%	52%
Specific fuel consumption	g/liter	48	48	57	51.00
Temp-corrected specific consumption	g/liter	50	50	59	53.00
Temp-corrected specific energy cons.	kJ/liter	763	775	902	813.33
Firepower	watts	2,251	3,004	2,333	2529.33

2. HIGH POWER TEST (HOT START)	units	Test 1	Test 2	Test 3	Average
Time to boil Pot # 1	min	16	19	18	17.67
Temp-corrected time to boil Pot # 1	min	17	20	18	18.33
Burning rate	g/min	15	12	13	13.33
Thermal efficiency	%	51%	57%	52%	53%
Specific fuel consumption	g/liter	50	46	49	48.33
Temp-corrected specific consumption	g/liter	52	48	50	50.00
Temp-corrected specific energy cons.	kJ/liter	804	735	770	769.67
Firepower	watts	3,883	2,979	3,358	3406.67

IWA PERFORMANCE METRICS	units	Test 1	Test 2	Test 3	Average
High Power Thermal Efficiency	%	53.40%	54.00%	50.10%	52.50%

IWA PERFORMANCE TIERS	Tier
High Power Thermal Efficiency	4

The average thermal efficiency of Fuelnzl Star-25 stove from the above tests was found to be **52.50%**.

Concluding Remarks:

The testing of the mentioned natural draft biomass cookstove was performed at laboratory of School of Energy Science & Engineering, IIT Guwahati, during November 2024, under supervision of Dr. Lepakshi Barbora and Mr. Debarshi Baruah of School of Energy Science & Engineering, IIT Guwahati. The observations and results pertain only to the particular equipment and product whose details are mentioned above. An uncertainty of $\pm 3\%$ may be considered in the results due to possible errors in human handling. The Investigators and School of Energy Science & Engineering, IIT Guwahati, does not accept any liability for any consequences including commercial or otherwise arising out of the utilization of the information contained in this report.



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